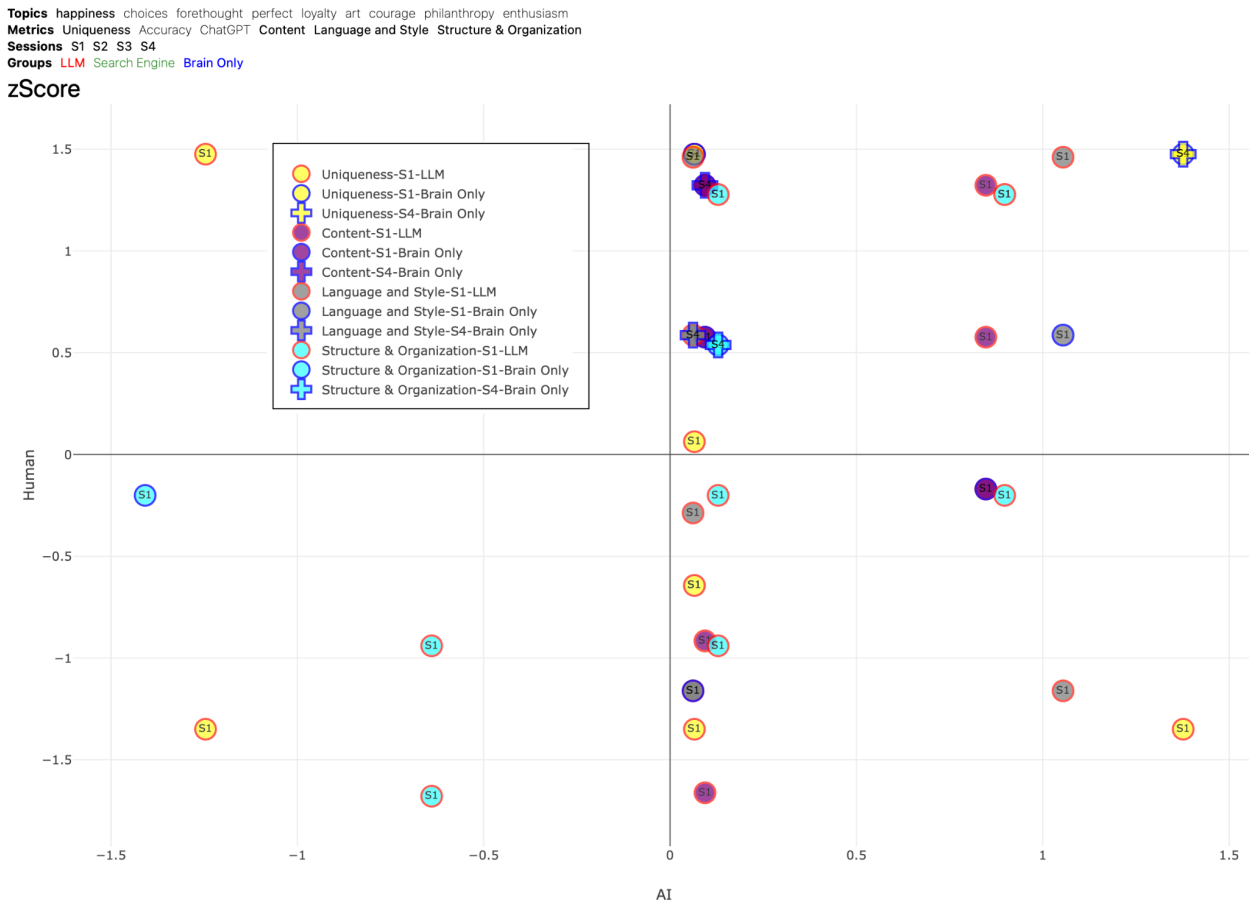


Figure 52. Z-Score Distribution of Assessments for topic HAPPINESS. This scatter plot displays the distribution of z-scores for AI judges and human evaluations, with human z-scores represented on the y-axis and AI z-scores on the x-axis. The plot offers a direct comparison of the variability in the way AI and human evaluators rate the essays across different metrics. Shapes represent different sessions, like circle is session 1, square is session 2, diamond is session 3, cross is session 4. Different fill colors represent different metrics, like yellow is uniqueness, purple is content, gray is language and style, cyan is structure and organization. Border color of each shape represents the group, like red is LLM group, Search Engine is green, and Brain-only is blue.

In topic LOYALTY (Figure 53) we can observe similar patterns to the previous figure, however this time AI judge score LLM uniqueness (yellow circle with red border) high (right side), while human teachers disagree by scoring it low (bottom).

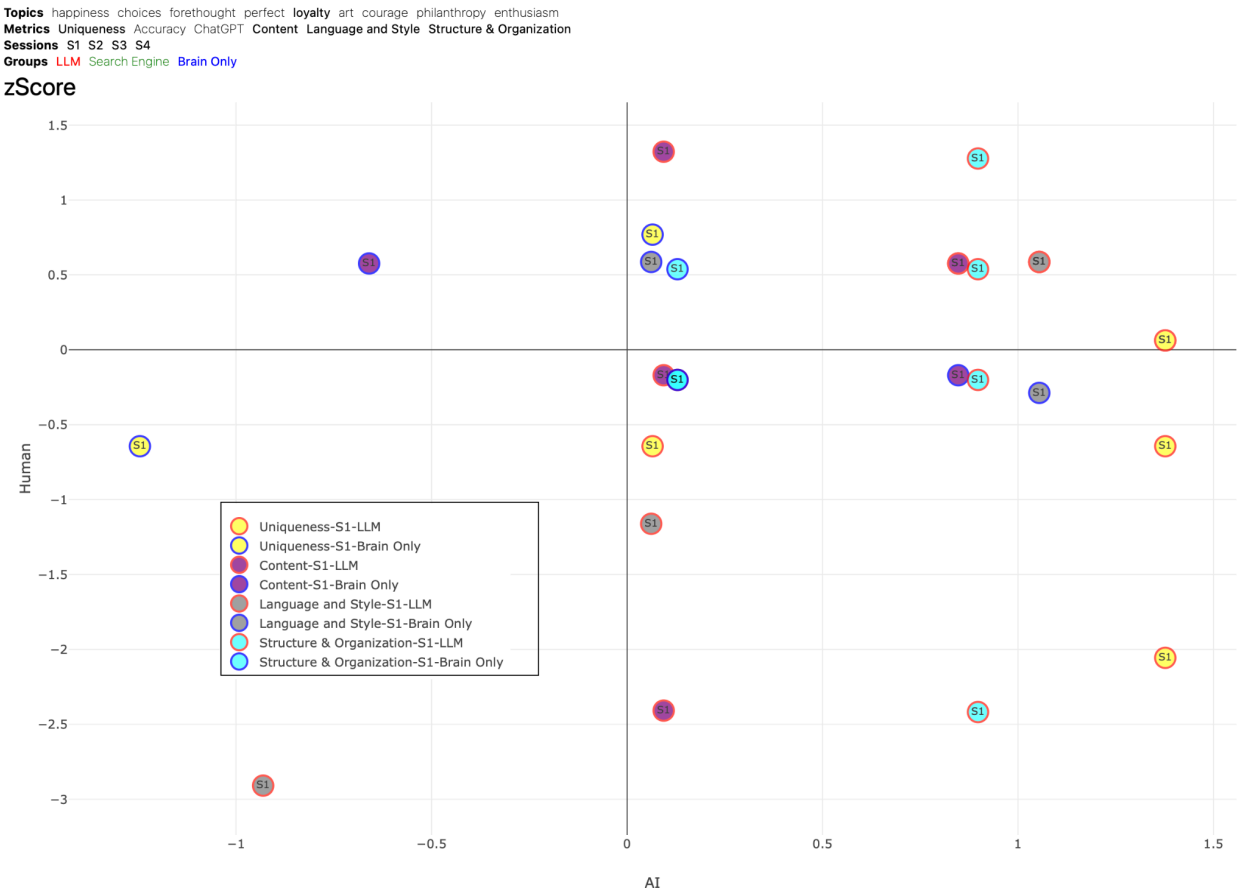


Figure 53. Z-Score Distribution of Assessments for topic LOYALTY. This scatter plot displays the distribution of z-scores for AI judges and human evaluations, with human z-scores represented on the y-axis and AI z-scores on the x-axis. The plot offers a direct comparison of the variability in the way AI and human evaluators rate the essays across different metrics. Shapes represent different sessions, like circle is session 1, square is session 2, diamond is session 3, cross is session 4. Different fill colors represent different metrics, like yellow is uniqueness, purple is content, gray is language and style, cyan is structure and organization. Border color of each shape represents the group, like red is LLM group, Search Engine is green, and Brain-only is blue.

For the PERFECT topic in Figure 54 we can observe metric structure and organization scored high by human teachers for the LLM group (cyan diamonds with red border), while Brain-only group (cyan diamonds with blue border) is ranked slightly above the mean, or almost always below it.

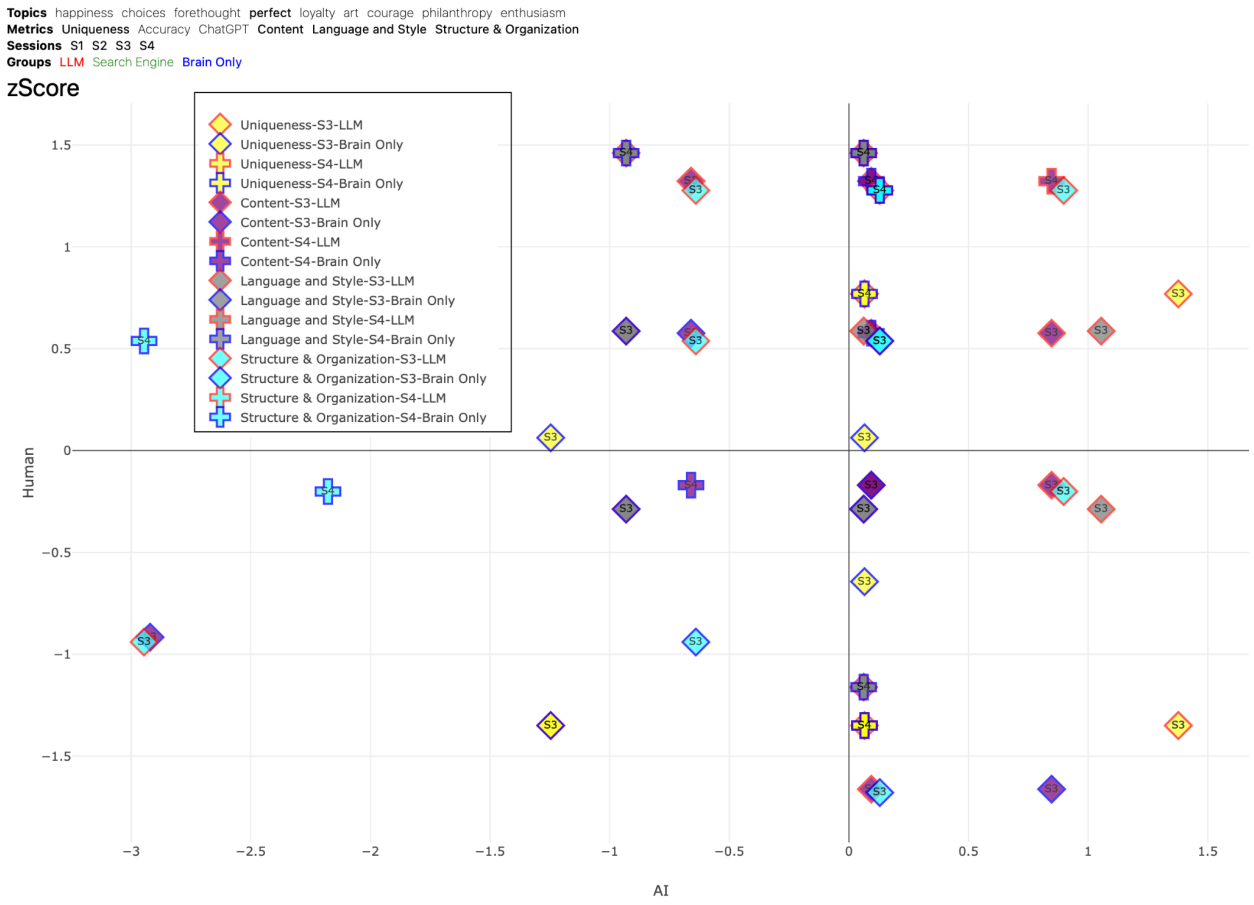


Figure 54. Z-Score Distribution of Assessments for topic PERFECT. This scatter plot displays the distribution of z-scores for AI judges and human evaluations, with human z-scores represented on the y-axis and AI z-scores on the x-axis. The plot offers a direct comparison of the variability in the way AI and human evaluators rate the essays across different metrics. Shapes represent different sessions, like circle is session 1, square is session 2, diamond is session 3, cross is session 4. Different fill colors represent different metrics, like yellow is uniqueness, purple is content, gray is language and style, cyan is structure and organization. Border color of each shape represents the group, like red is LLM group, Search Engine is green, and Brain-only is blue.

In the topic PHILANTHROPY (Figure 55) structure and organization metric almost always scored high by the AI judge for Brain-only group (cyan squares with blue border, right side), while only on essay reached positive score by both AI judge and human teachers for LLM group (cyan square with red border, top right quadrant) while the rest were scored below the mean by both groups (bottom left quadrant).

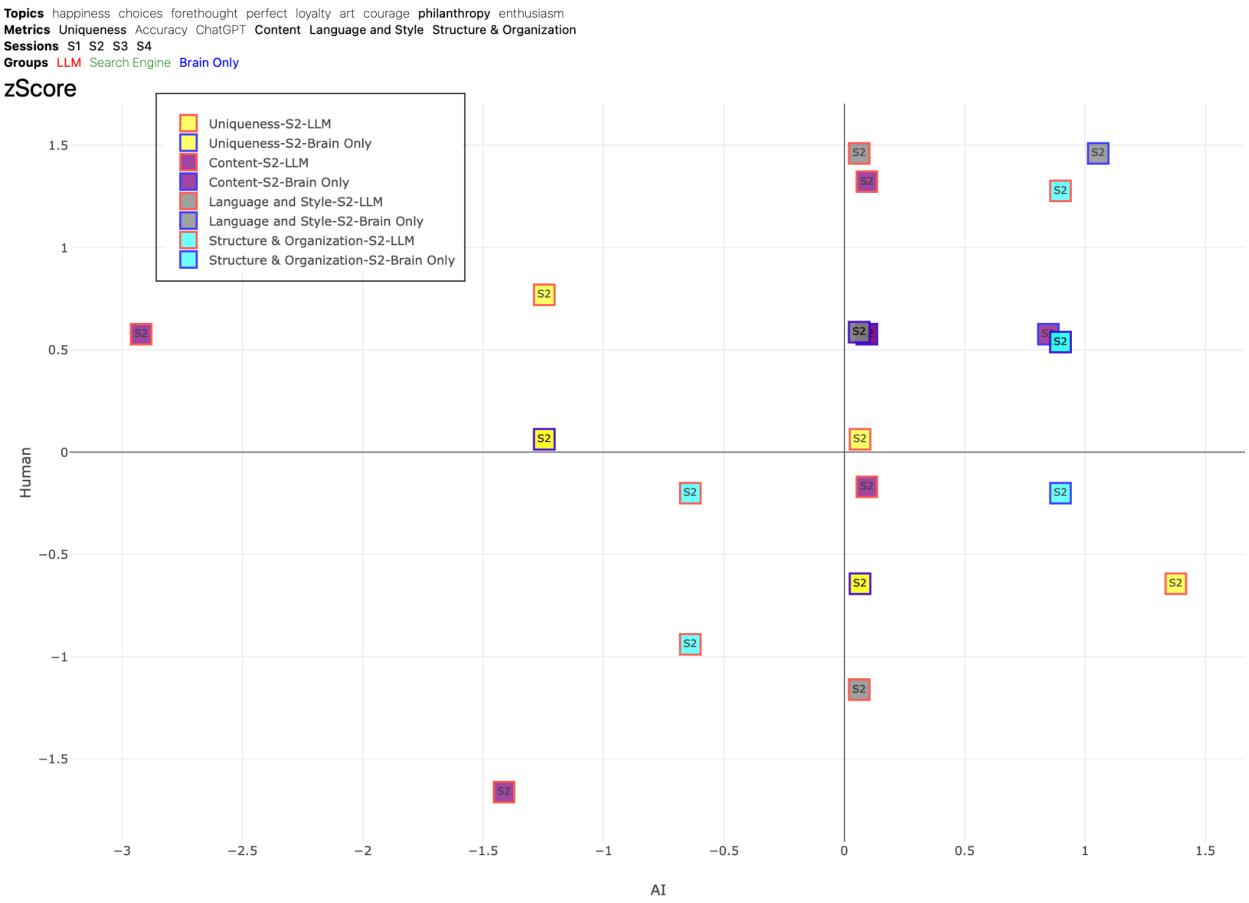


Figure 55. Z-Score Distribution of Assessments for topic PHILANTHROPY. This scatter plot displays the distribution of z-scores for AI judges and human evaluations, with human z-scores represented on the y-axis and AI z-scores on the x-axis. The plot offers a direct comparison of the variability in the way AI and human evaluators rate the essays across different metrics. Shapes represent different sessions, like circle is session 1, square is session 2, diamond is session 3, cross is session 4. Different fill colors represent different metrics, like yellow is uniqueness, purple is content, gray is language and style, cyan is structure and organization. Border color of each shape represents the group, like red is LLM group, Search Engine is green, and Brain-only is blue.

Interviews

At the end of each session we conducted an interview with each participant.

To analyze these post-assessment interviews, we used the PaCMAP clustering to extract the contextual proximity of 15 clusters. Main clusters are visualized in Figure 56. List of clusters' descriptions is available in Appendix A.

In the interviews we conducted after the participants finished the sessions, we found several interesting patterns. For example, cluster 7 shows balanced agreement on the importance of thinking before speaking. Cluster 3 consists of complaints of the Brain-only group about limited time and their potential inability to deliver satisfactory results. Cluster 10 shows how valuable it was for participants to own their ideas during writing the essays, and we can see the cluster of small blue dots (LLM) shows participants' realization. Cluster 15 shows popularity of using ChatGPT to generate the intro, and mostly in sessions 1, 2, 3, and almost not in session 4.

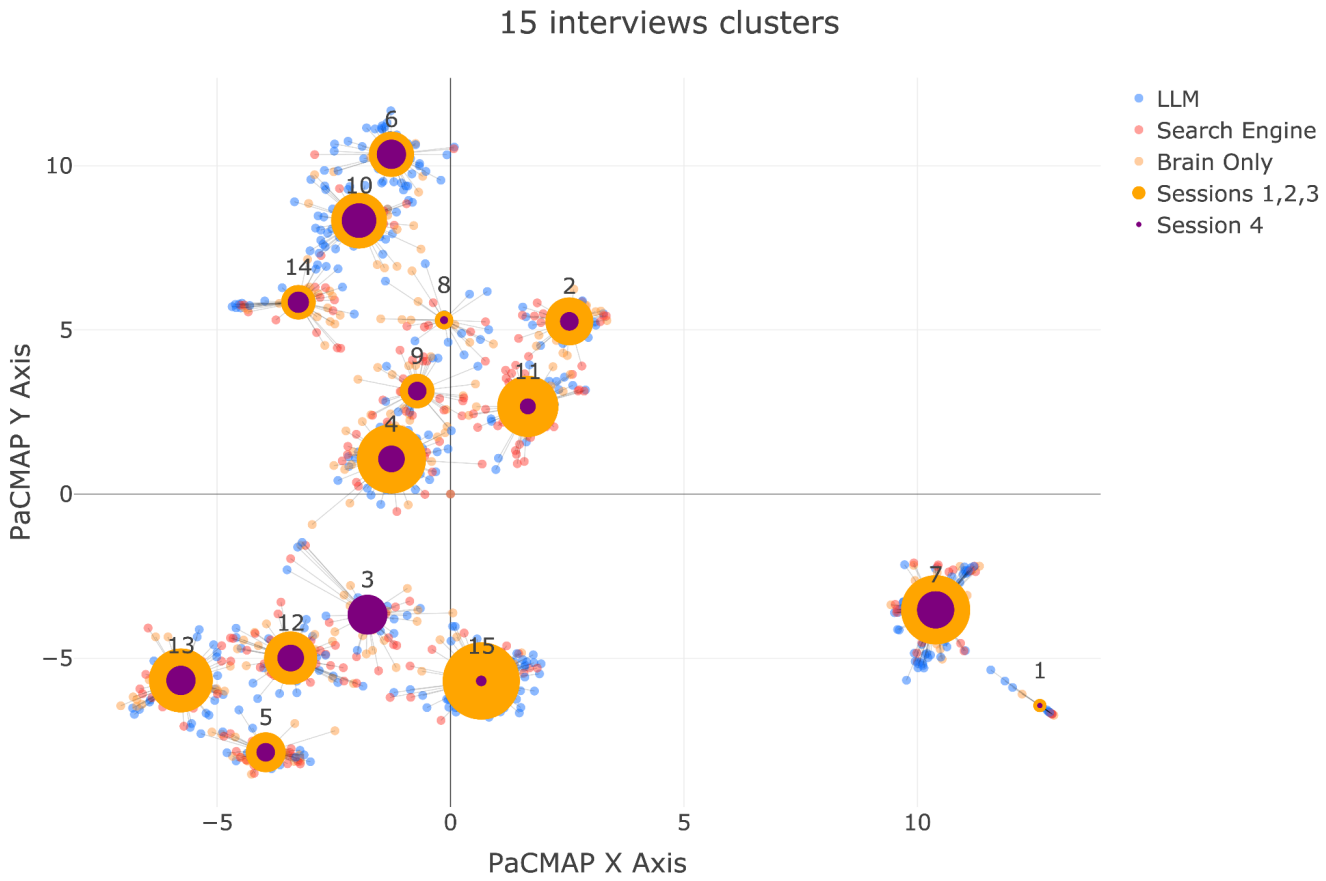


Figure 56. PaCMAP defined clusters of the interview insights between session 4 and sessions 1, 2, 3. See the insights in the appendix quoted below for each cluster on the map, top to bottom, left to right. List of descriptions is available in Appendix A.

EEG ANALYSIS

We have collected EEG data using Enobio headset with 32 electrodes [128]: P7, P4, Cz, Pz, P3, P8, O1, O2, T8, F8, C4, F4, Fp2, Fz, C3, F3, Fp1, T7, F7, Oz, PO4, FC6, FC2, AF4, CP6, CP2, CP1, CP5, FC1, FC5, AF3, PO3 at 500 samples per second (SPS) at 24 bit resolution with measurement noise less than 1 μV root mean square (RMS). We applied a low pass filter at 0.1 Hz and high pass filter at 100 Hz, and notch filter at 60 Hz (power line frequency in the U.S.). We applied Independent Component Analysis (ICA) to each session and visually inspected correlating eye blink activity, and excluded those components from the regenerated filtered EEG signal. We normalize the signal using the min-max scaling method converting all values to the range from 0 to 1.

Frequency bands were defined as follows: the Delta band spanned 0.1-4 Hz and was further subdivided into low-delta (0.2-0.83 Hz), mid-delta (0.83-2.66 Hz), and high-delta (2.66-4 Hz) sub-bands [69]. Theta activity encompassed 4-8 Hz. The Alpha band covered 8-12 Hz, with low-alpha defined as 8-10 Hz and high-alpha as 10-12 Hz. Beta band extended from 12-30 Hz and was subdivided into low-beta (12-15 Hz), mid-beta (15-18 Hz), and high-beta (18-30 Hz). Finally, Gamma activity ranged from 30-100 Hz, including low-gamma (30-44 Hz) and high-gamma (44-100 Hz) components.

In this paper we only report our results for low/high alpha, low/high beta, low/high delta and theta bands.

Dynamic Directed Transfer Function (dDTF)

In this work we decided to compare how different parts of brain surface influence others in the context of different tasks performed within different groups across the sessions, including the switch sessions (4) at the end, we analyzed them across the bands and subbands, and we can see the results of this analysis in the next section.

dDTF [70] is a method derived from DTF that focuses on the dynamic fitting of Multivariate Autoregressive (MVAR) models to find the most effective connectivity in a frequency domain of EEG, in this study we conducted pairwise analysis of channels (electrodes), and unlike a coherence method, the calculated data is not symmetric (meaning A→B is not equal B→A).

Before the dDTF calculation we first have to calculate the MVAR model for each pair. MVAR models defined with the window size [71] and the order [72]. It will be used later in Granger Causality [73] to estimate the effectiveness of the connectivity. Given the hardware (32 channels Enobio by Neuroelectrics) used to record EEG data sampled at 500 Hz, we evaluated different window sizes from 0.5 seconds to 1 minute and ended up using 1 second window size since it gave best performance and accuracy. Since Joint Order and Coefficient Estimation (JOCE) and Least Absolute Shrinkage Selection Operator (LASSO) [71] were more compute heavy for the amount of data collected, we opted to use conventional Akaike's information criterion (AIC) and Bayesian information criterion (BIC) for model order validation dynamically for each pair over the

time of 20 minutes during the essay writing task, due to complexity of the data and its volume the AIC implementation included logarithm of the determinant of a positive-definite matrix using its Cholesky decomposition to give faster and more error-prone precision.

MVARs are typically calculated using the following equation:

$$X(t) = \sum_{p=1}^{p} A_p X(t-p) + E(t) \quad (6)$$

Where $X(t)$ is an EEG signal, P is the model order, A_p is a coefficient matrix, which is used later for dDTF, $E(t)$ is a residual gaussian noise with a covariance matrix.

Both AIC and BIC suggested model order to be on the lower end of 5, while about 25% of pairs across all sessions were in the 6-8 order range, rarely going up to 10.

The dDTF will require transfer function in frequency domain via Fast Fourier Transform (FFT), DTF, and Partial Coherence. Transfer function looks like this:

$$H(f) = (\sum_{p=0}^{p} A_p e^{-j2\pi fp}) \quad (7)$$

Directed Transfer Function:

$$DTF_{ij}(f) = \frac{|H_{ij}(f)|}{\sqrt{\sum_{k=1}^{K} |H_{ik}(f)|^2}} \quad (8)$$

Partial Coherence:

$$\gamma_{ij}^2(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f)S_{jj}(f)} \quad (9)$$

Final dDTF [74]:

$$dDTF_{ij}(f) = DTF_{ij}(f) \cdot \sqrt{\gamma_{ij}^2(f)} \quad (10)$$

To normalize the data we divided each computed dDTF for a particular frequency by the quadratic sum of the EEG values in row j. Which were finally accumulated into the dDTF value per band using frequency band ranges described in the beginning of this section.

We also examined full frequency DTF (ffDTF) which has embedded Granger Causality effect due to the full spectrum aspect, but unfortunately the nature of the task performed required multi-band analysis to better demonstrate the differences between the different groups of participants therefore we decided to only used dDTF in this work.

We did experiment with different orders and window sizes for the MVAR model to drive different results in the dDTF, but AIC yielded that for our data and window size and minimal order of 5 gave the best results for multi-band and subband dDTF effective connectivity.

For all the sessions we calculated dDTF for all pairs of electrodes $32 \times 31 = 992$ and ran repeated measures analysis of variance (rmANOVA) within the participant and between the participants within the groups. Due to complexity of the data and volume of the collected data

we ran rmANOVA ≤ 1000 times each. To denote different levels of significance in figures and results, we adopted the following convention:

- $p < 0.05$ was considered statistically significant and is marked with a single asterisk (*)
- $p < 0.01$ with a double asterisk (**)
- $p < 0.001$ with a triple asterisk (***)

This notation is used consistently throughout all figures and tables. When multiple comparisons were involved, p-values were adjusted using the False Discovery Rate (FDR) correction, and the significance markers refer to the adjusted values unless stated otherwise.

EEG Results: LLM Group vs Brain-only Group

Alpha Band Connectivity

The most pronounced difference emerged in alpha band connectivity, with the Brain-only group showing significantly stronger semantic processing networks. The connection from left parietal (P7) to right temporal (T8) regions demonstrated highly significant group differences (p=0.0002, dDTF: Brain-only group=0.053, LLM group=0.009). This P7→T8 pathway was complemented by enhanced connectivity from parieto-occipital regions to anterior frontal areas (PO4→AF3: p=0.0025, Brain-only group=0.024, LLM group=0.009). The temporal region T8 emerged as a major convergence hub in the Brain-only group (Figure 57, Appendix F, L, I).

The Brain-only group also demonstrated stronger occipital-to-frontal information flow (Oz→Fz: p=0.003, Brain-only group=0.02, LLM group=0.1). The total significant connectivity for the Brain-only group was equal to 79 connections compared to only 42 connections for the LLM group.

Alpha band connectivity is often associated with internal attention and semantic processing during creative ideation [75]. The higher alpha connectivity in the Brain-only group suggests that writing without assistance most likely induced greater internally driven processing, consistent with the idea that these participants had to generate and combine ideas from memory without external cues. In fact, creativity research shows that alpha activity (especially in upper-alpha) increases with internal semantic search and creative demand at frontal and parietal sites [75]. Brain-only group's elevated fronto-parietal alpha connectivity aligns with this finding: their brains likely engaged in more internal brainstorming and semantic retrieval. The LLM group, taking into account the LLM's suggestions, may have relied less on purely internal semantic generation, leading to lower alpha connectivity, because some creative burden was offloaded to the tool.

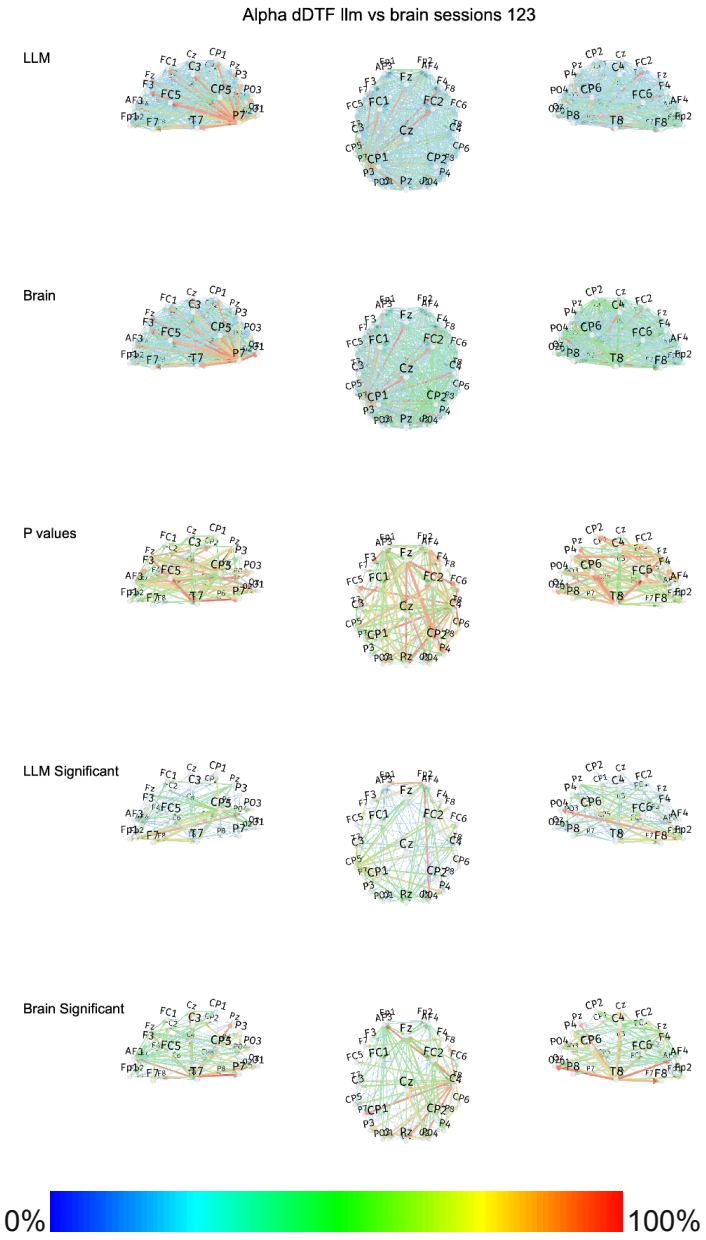


Figure 57. Dynamic Direct Transfer Function (dDTF) for Alpha band between LLM and Brain groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Beta Band Connectivity

Beta band analysis revealed contrasting patterns between low and high-beta frequencies. In low-beta (13-20 Hz), Brain-only group maintained slight superiority (total connectivity: 2.854 vs 2.653), with particularly strong temporal-to-frontal connections (P7→T8: p=0.0003, dDTF: Brain-only group=0.057, LLM group=0.009). However, high-beta (20-30 Hz) showed more balanced connectivity patterns with the LLM group demonstrating stronger cognitive control networks. Within the right hemisphere, the Brain-only group also tended toward stronger frontal→temporal beta connectivity (e.g. right frontal to right temporal lobe) (Figure 58, Appendix G, M, J). The LLM group did not show increases in any beta connections relative to the Brain-only group, rather, all major beta band connections were either stronger in the Brain-only group or similar between groups. This suggests a broad enhancement of beta-range coupling in the brain-only condition.

Beta band connectivity is often linked to active cognitive processing, focused attention, and sensorimotor integration. The higher beta connectivity in the Brain-only group likely reflects their sustained cognitive and motor engagement in composing their essays without external tools. Writing without a tool meant the Brain-only group had to continuously generate text and maintain their plan, which engaged executive functions and possibly the motor planning for typing, processes known to manifest in beta oscillations.

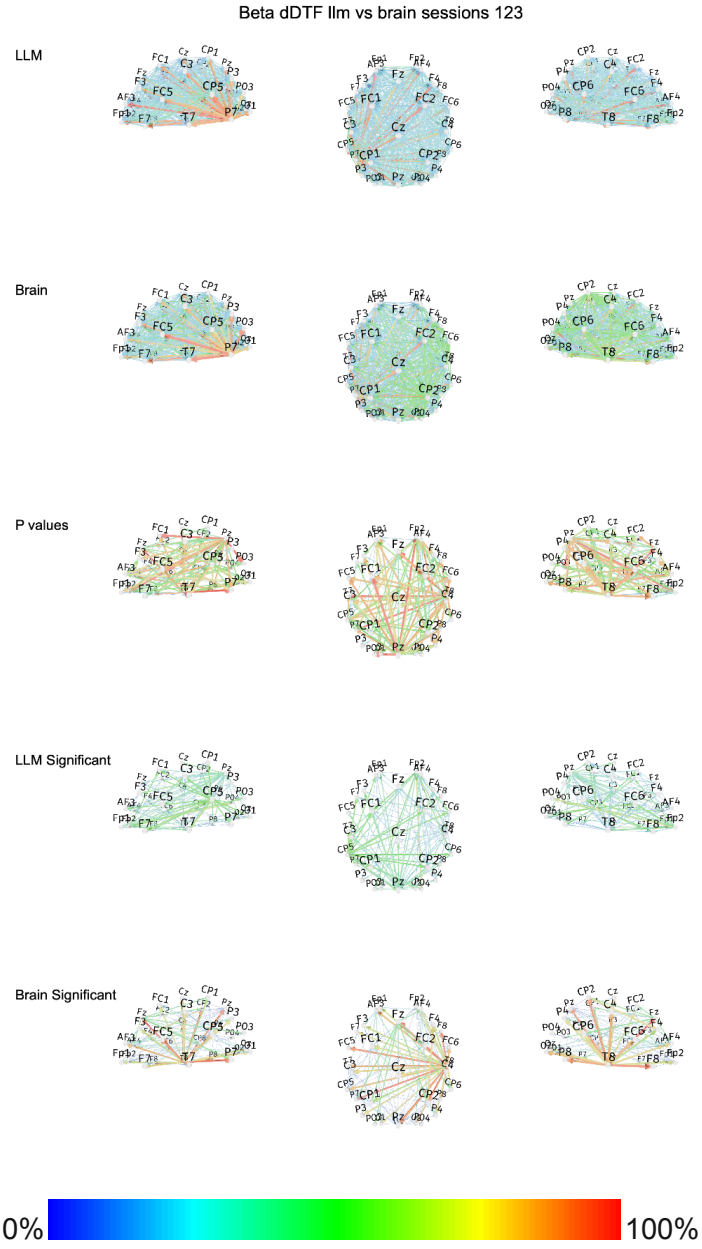


Figure 58. Dynamic Direct Transfer Function (dDTF) for Beta band between LLM and Brain groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Delta Band Connectivity

Delta band analysis revealed Brain-only group's dominance in executive monitoring networks. The most significant connection was from left temporal to anterior frontal regions (T7→AF3: p=0.0002, dDTF: Brain-only group=0.022, LLM group=0.007), indicating enhanced executive control engagement (Figure 59, Appendix H, N, K). This was supported by additional connections converging on AF3 from multiple regions (FC6→AF3: p=0.0007, F3→AF3: p=0.0020 and many others).

The anterior frontal region AF3 served as a major convergence hub in the Brain-only group. The Brain-only group demonstrated a clear superiority with 78 connections showing the Brain-only group compared to only 31 in the opposite direction. Additionally, the Brain-only group showed stronger inter-hemispheric delta connectivity between frontal areas, consistent with more coordinated low-frequency activity across hemispheres during unassisted writing [76] .

Delta band connectivity is thought to reflect broad, large-scale cortical integration and may relate to high-level attention and monitoring processes even during active tasks. In the creative writing context, significant delta band connectivity differences likely point to greater recruitment of distributed neural networks when writing without external aid. Prior studies of creative writing stages found that delta band effective connectivity can increase when moving from an exploratory stage to an intense generation stage [76]. The higher delta connectivity in the Brain-only group could indicate that these participants engaged more multisensory integration and memory-related processing while formulating their essays. Another perspective is that delta oscillations sometimes relate to the default mode during tasks, Brain-only group's higher delta might reflect deeper immersion in internally-driven thought (since they must self-generate content), whereas LLM group's participants thought process could be intermittently interrupted or guided by suggestions from the LLM, potentially dampening sustained delta connectivity.

To summarize, the delta-band differences suggest that unassisted writing engages more widespread, slow integrative brain processes, whereas assisted writing involves a more narrow or externally anchored engagement, requiring less delta-mediated integration.

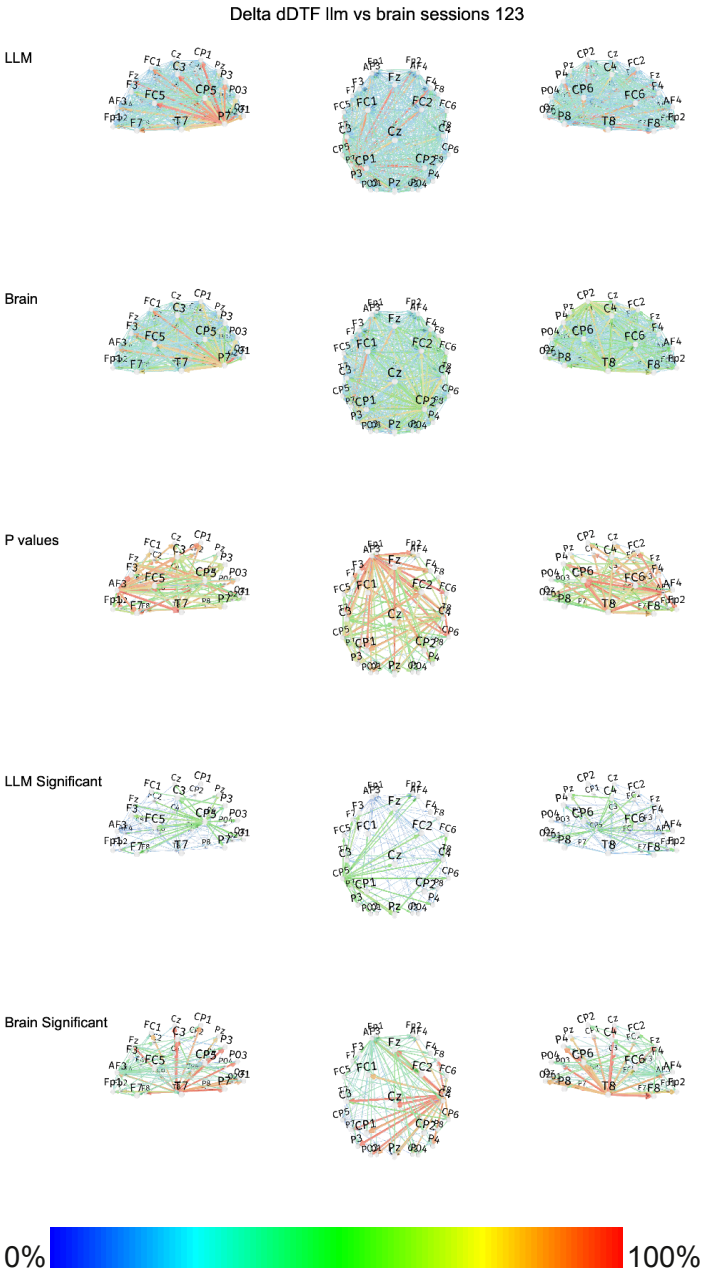


Figure 59. Dynamic Direct Transfer Function (dDTF) for Delta band between LLM and Brain groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Theta Band Connectivity

Theta band connectivity patterns were significant in the Brain-only group. The most significant connection was from the parietal midline to the right temporal regions (Pz→T8: p=0.0012, dDTF: Brain-only group=0.041, LLM group=0.009). Additional significant connections included occipital-to-frontal pathways (Oz→Fz: p=0.0016) and fronto-central to anterior frontal connections (FC6→AF3: p=0.0017).

The anterior frontal region AF3 again emerged as a convergence hub in the Brain-only group. The overall pattern showed 65 connections for the Brain-only group versus 29 for the LLM group (Figure 60, Appendix O), indicating more extensive theta-band processing in tool-free writing.

Theta band differences were most apparent in networks involving frontal-midline regions and posterior regions. Brain-only group displayed significantly stronger frontal → posterior theta connectivity, especially from midline prefrontal areas (e.g. Fz or adjacent frontal leads) toward parietal and occipital areas. In addition, inter-hemispheric theta connectivity (frontal-frontal across hemispheres) was elevated in the Brain-only group. These patterns align with a scenario where the frontal cortex of the Brain-only group served as a hub driving other regions in the theta band. In contrast, LLM group had uniformly lower theta directed influence; notably, fronto-parietal theta connections that were prominent in Brain-only group were relatively weak or absent in LLM group. No theta band connection showed higher strength in the LLM group than in the Brain-only group. The overall theta network thus appears more active and directed from frontal regions in non-assisted writing.

Theta band activity is closely linked to working memory load and executive control. In fact, frontal theta power and connectivity increase linearly with the demands on working memory and cognitive control [77]. The much higher theta connectivity in the Brain-only group strongly suggests that writing without assistance placed a greater cognitive load on participants, engaging their central executive processes. Frontal-midline theta is known as a signature of mental effort and concentration, often arising from the need to hold and manipulate information in mind [77]. Brain-only group's brain activity exhibited more intense frontal theta networking (frontal regions driving other areas), indicating they were most likely actively coordinating multiple cognitive components (ideas, linguistic structures, attention) in real-time to compose their essays. This finding aligns with the expectation that executive function was more heavily involved in the absence of any tools. The LLM group, by contrast, had significantly lower theta connectivity, consistent with a reduced working memory burden: the LLM likely provided suggestions that lessened the need for participants to internally generate and juggle as much information. In other words, the LLM group did not need to sustain as much frontal theta-driven coordination, because the external aid helped scaffold the writing process. The theta results thus highlight that non-assisted writing invoked greater engagement of the brain's executive control network, whereas tool-assisted writing allowed for a lower load. This may have freed cognitive resources for other aspects (like evaluating the tool's output), but it clearly diminished the need for intense theta-mediated integration.

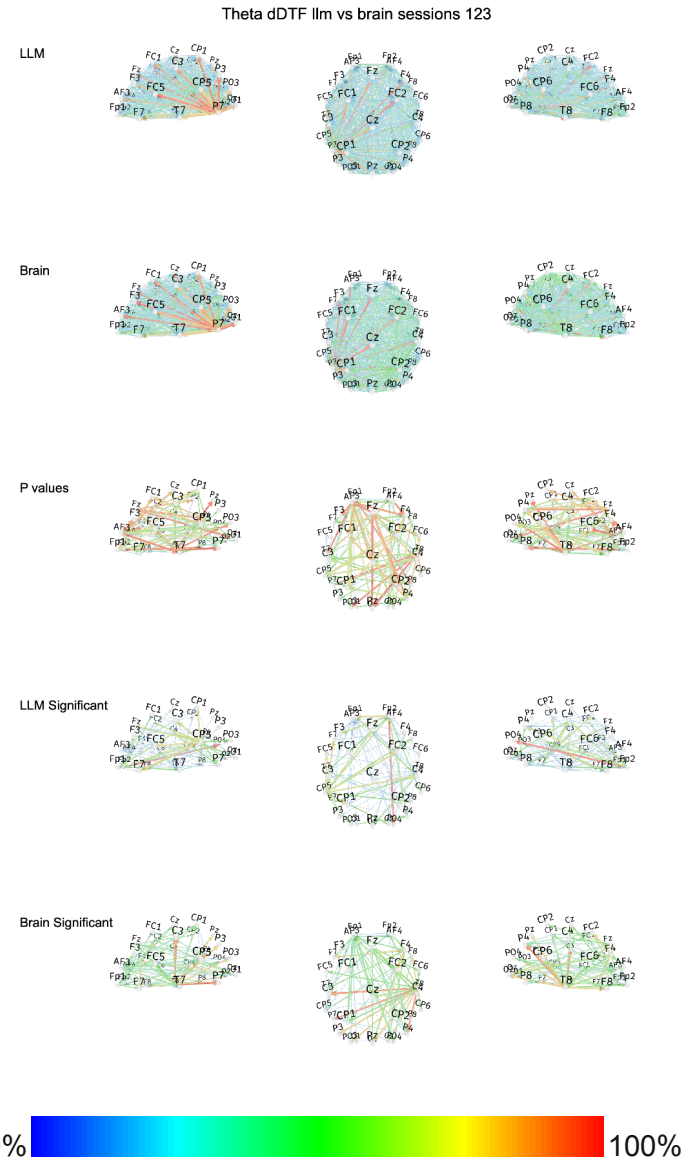


Figure 60. Dynamic Direct Transfer Function (dDTF) for Theta band between LLM and Brain groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Summary

Our findings offer an interesting glimpse into how **LLM-assisted vs. unassisted writing** engaged the brain differently. In summary, writing an essay *without* assistance (Brain-only group) led to **stronger neural connectivity**, with particularly large increases in the **theta and high-alpha bands**. This indicates that participants in the Brain-only group had to heavily engage their own cognitive resources: frontal executive regions orchestrated more widespread communication with other cortical areas (especially in the theta band) to meet the high working memory and planning demands of formulating their essays from scratch. The elevated theta connectivity, centered on frontal-to-posterior directions, often represents increased cognitive load and executive control [77]. In parallel, the Brain-only group exhibited enhanced high-alpha connectivity in fronto-parietal networks, reflecting the internal focus and semantic memory retrieval required for creative ideation without external aid [75].

The delta band differences revealed that the Brain-only group also engaged more large-scale integrative processes at slow frequencies, possibly reflecting deeper encoding of context and an ongoing integration of non-verbal memory and emotional content into their writing [76]. Tools-free writing activated a **broad spectrum** of brain networks, from slow to fast rhythms, indicating a **holistic cognitive workload**: memory search, idea generation, language formulation, and continuous self-monitoring were all in play and coordinated by frontal executive regions.

In contrast, **LLM-assisted writing (LLM group)** elicited a generally **lower connectivity profile**. While the LLM group certainly engaged brain networks to write, the presence of a LLM appears to have **attenuated the intensity and scope of neural communication**. The significantly lower frontal theta connectivity in the LLM group possibly indicates that their **working memory and executive demands were lighter**, presumably because the bot provided external cognitive support (e.g. suggesting text, providing information, structure). Essentially, some of the 'human thinking' and planning was offloaded, and the brain did not need to synchronize as extensively at theta frequencies to maintain the writing plan. LLM group's reduced beta connectivity possibly indicated a somewhat lesser degree of sustained concentration and arousal, aligning with a potentially lower effort during writing.

Another interesting insight is the **difference in information flow directionality** between the groups. Brain-only group showed evidence of greater **bottom-up flows** (e.g. from temporal/parietal regions to frontal cortex) during essay writing. This bottom-up influence can be interpreted as the brain's semantic and sensory regions 'feeding' novel ideas and linguistic content into the frontal executive system, essentially the brain generating content internally and the frontal lobe integrating and making decisions to express it [76]. In contrast, LLM group, with external input from the bot, likely experienced more **top-down directed connectivity** (frontal → posterior in high-beta). Their frontal cortex was often in the role of integrating and filtering the tool's contributions (an external source), then imposing it onto their overall narrative. This might be to an extent analogous to a 'preparation' phase in creative tasks where external stimuli are interpreted by frontal regions sending information to posterior areas [76]. Our results support this: LLM group had relatively higher frontal → posterior connectivity than Brain-only group in

some bands (notably in beta and high-beta), consistent with tool-related top-down integration, whereas Brain-only group had higher posterior → frontal flows (as seen in delta band results and overall patterns) consistent with self-driven idea generation [76].

From a cognitive load perspective, the neural connectivity metrics align well with expectations. Non-assisted writing is a high-load task, the brain must handle idea generation, organization, composition, all internally, and indeed Brain-only group's connectivity profile (high frontal theta, broad network activation) is typical of a high mental workload state [77, 78]. Tool assistance, on the other hand, distributed some of that load outward, resulting in a lower connectivity demand on the brain's networks (especially the frontally-mediated networks for working memory). Interestingly, while this made the task possibly easier (lower load), it also seems to correlate with lower alpha connectivity, which is prominent in creativity tasks, suggesting a potential trade-off: the LLM might streamline the process, but the user's brain may engage less deeply in the creative process.

Regarding executive function, the results show Brain-only group's prefrontal cortex was highly involved as a central hub (driving strong theta and beta connectivity to other regions), indicating substantial executive control over the writing process. LLM group's prefrontal engagement was comparatively lower, implying that some executive functions (like maintaining context, planning sentences) were most likely partially taken over by the LLM's automation. However, the LLM group still needed executive oversight to evaluate and integrate LLM suggestions, which is reflected in the top-down connectivity they exhibited. So, while the **quantity** of executive involvement was less for LLM users, the **nature** of executive tasks may have shifted, **from generating content to supervising the AI-generated content.**

In terms of **creativity**, one could argue that Brain-only group's brain networks were more activated in the manner of creative cognition: their enhanced fronto-parietal alpha connectivity suggest rich internal ideation, associative thinking, and possibly engagement of the default-mode network to draw upon personal ideas and memory [75]. LLM group's reduced alpha connectivity and increased external focus might indicate a more convergent thinking style, they might lean on the LLM's suggestions (which could constrain the range of ideas) and then apply their judgment, rather than internally diverging to a wide space of ideas.

In conclusion, the directed connectivity analysis reveals a clear pattern: **writing without assistance increased brain network interactions across multiple frequency bands**, engaging higher cognitive load, stronger executive control, and deeper creative processing. **Writing with AI assistance, in contrast, reduces overall neural connectivity**, and shifts the dynamics of information flow. In practical terms, a LLM might free up mental resources and make the task feel easier, yet the brain of the user of the LLM might not go as deeply into the rich associative processes that unassisted creative writing entails.

EEG Results: Search Engine Group vs Brain-only Group

Alpha Band Connectivity

In the alpha band, the Brain-only group exhibited stronger overall brain connectivity than the Search Engine group (Figure 61, Appendix Z, AC, AF). The dDTF values across significant connections were higher for the Brain-only group (0.423) compared to the Search Engine group (0.288). This indicates more robust alpha-band coupling when participants wrote without external aids. Directionality-wise, the Brain-only group showed greater outgoing influences from posterior regions (e.g. right occipital O2, left temporal T7, occipital Oz) and stronger incoming influences to the right frontal cortex (F4). In fact, F4 emerged as a major sink in Brain-only group's alpha network, receiving six significant connections (total incoming dDTF ~0.203 vs. 0.074 in Search Engine group). By contrast, Search Engine group showed modestly more alpha outputs from a few sites (e.g. left occipital O1, parieto-occipital PO4) and slightly greater inputs to frontopolar Fp2 and midline Cz, but these were fewer and weaker than Brain-only group's frontal hub pattern.

Several specific alpha band connections were significantly stronger in the Brain-only group. For instance, FC5→T8, F4→PO3, and T7→T8 showed higher dDTF in the Brain-only group (indicating stronger directed influence from frontal/temporal sources to temporal/parietal targets). Several connections were stronger for Search Engine group, notably Fp1→Cz and posterior-to-frontal links like P4→Fp2 were higher for Search Engine group, but there were very few of these cases. All reported connections were statistically significant ($p < 0.05$), with the strongest differences reaching p ~0.01-0.02.

As we mentioned in the previous section of the paper, alpha band coherence is often associated with attentional control and internal information processing. The finding that the Brain-only group engaged more alpha connectivity (especially between posterior areas and frontal executive regions) suggests that writing without internet support required greater internal attention and memory integration. This resonates with prior studies showing that alpha band functional connectivity increases during high cognitive load and working memory demands in healthy individuals. Brain-only group's brain may have been synchronizing frontal and posterior regions to internally retrieve knowledge and organize the essay content. In contrast, Search Engine group's lower alpha connectivity (and fewer frontal hubs) might reflect reduced reliance on internal memory due to the availability of online information, consistent with the 'Google effect,' wherein easy access to external information can diminish the brain's tendency to internally store and connect information [37].

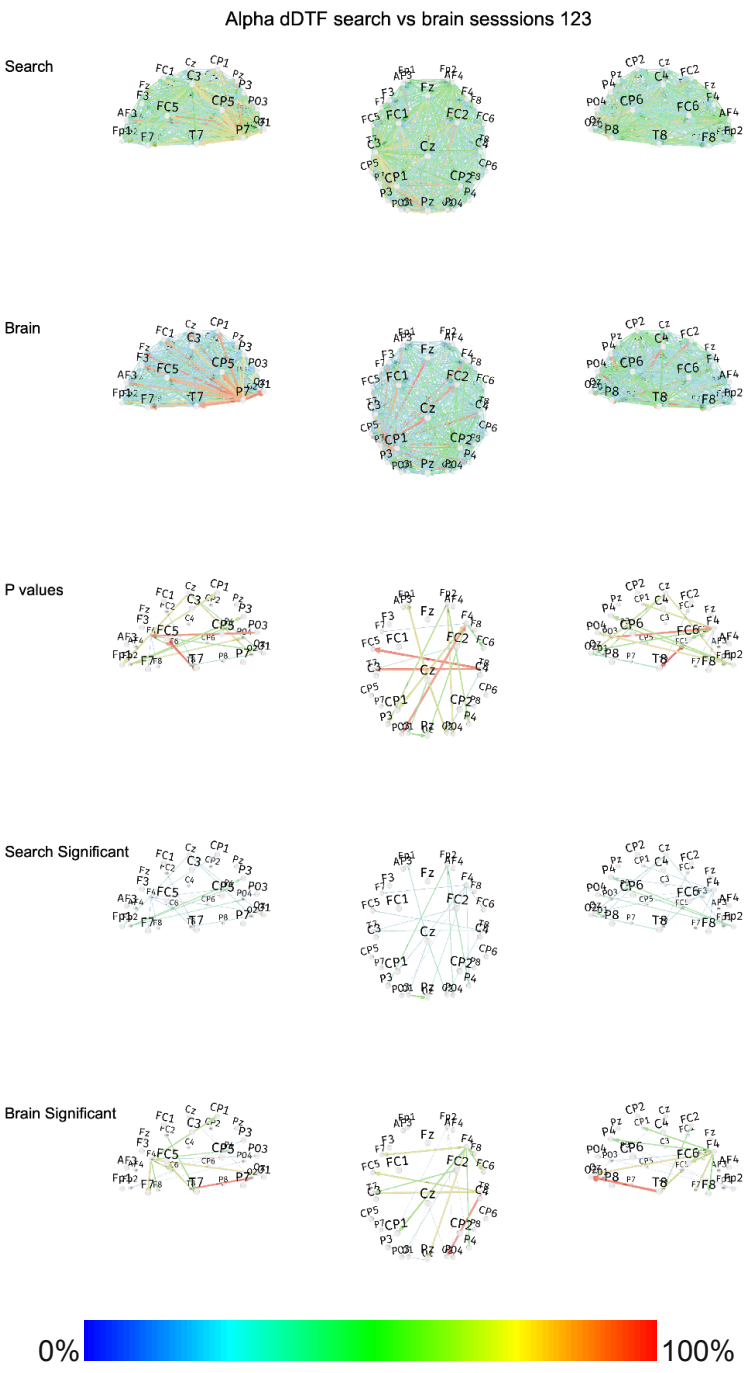


Figure 61. Dynamic Direct Transfer Function (dDTF) for Alpha band between Search Engine and Brain-only groups, only for sessions 1,2,3, excluding session 4. Rows 1 (Search Engine group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Beta Band Connectivity

Beta band connectivity displayed a more complex pattern. Brain-only group's total significant beta connectivity was slightly higher in magnitude (sum dDTF 0.417 for Brain-only group vs. 0.355 for Search Engine group), but Search Engine group showed a greater number of beta connections where it dominated (11 connections vs. 7 for Brain-only group). This suggests that while the Brain-only group had a slight edge in overall beta strength, the Search Engine group had numerous beta links (albeit some of smaller effect) in its favor.

Important differences were observed at the parietal midline (Pz), the Search Engine group had 7 significant inputs converging on Pz (total incoming beta 0.151) versus only 0.052 in the Brain-only group (Figure 62, Appendix AA, AD, AG). This indicates that with internet support, participants' brains funneled more beta-band influence into Pz (a region associated with visuo-spatial processing and integration). In contrast, the Brain-only group showed stronger beta inputs to the right temporal region (T8), 4 connections totaling 0.246 (vs. 0.085 in the Search Engine group). Brain-only group also had unique beta outputs from the left temporal cortex (T7) that were higher, specifically contributing to a robust T7→T8 connection (dDTF ~0.060 vs 0.022). Meanwhile, several fronto-parietal beta connections were stronger in the Search Engine group: for example, PO3→Pz, FC5→Pz, and Fp2→Pz (all projecting into the Pz hub) had larger dDTF in Search Engine group. These findings potentially indicate that Search Engine group's beta network centered on integrating externally gathered information (visual input, search engine results) in parietal regions, whereas Brain-only group's beta network engaged more bilateral communication involving temporal areas (possibly related to language and memory retrieval).

The strongest beta difference was F4→PO3 (right frontal to left parieto-occipital), highly significant, $p \approx 0.006$. Most other top beta differences were moderately significant (p ~0.02-0.04), and only connections with $p < 0.05$ were considered.

Beta band connectivity is commonly linked to active cognitive processing, sensorimotor functions, and top-down control [79]. The parietal beta connectivity in Search Engine group may reflect greater engagement with visual components of the search engine and motor aspects of the task: e.g. scrolling through online content could drive beta synchronization in visuo-motor networks (midline parietal and sensorimotor sites). This aligns with Search Engine showing beta activity increases during externally guided visual tasks [79] and during motor planning. On the other hand, Brain-only group's inclusion of temporal lobe in beta networks suggests deeper semantic or language processing, possibly formulating content from memory, engaging language networks. Such distributed beta connectivity might relate to the internal organization of knowledge and creative idea generation, processes that have been associated with beta oscillations in frontal-temporal regions [80]. In summary, internet-aided writing (Search Engine group) shifted beta band resources toward handling external information (visual attention, coordination of search engine and scrolling), whereas no-tools writing (Brain-only group) maintained beta connectivity more for internal information processing and cross-hemispheric communication.

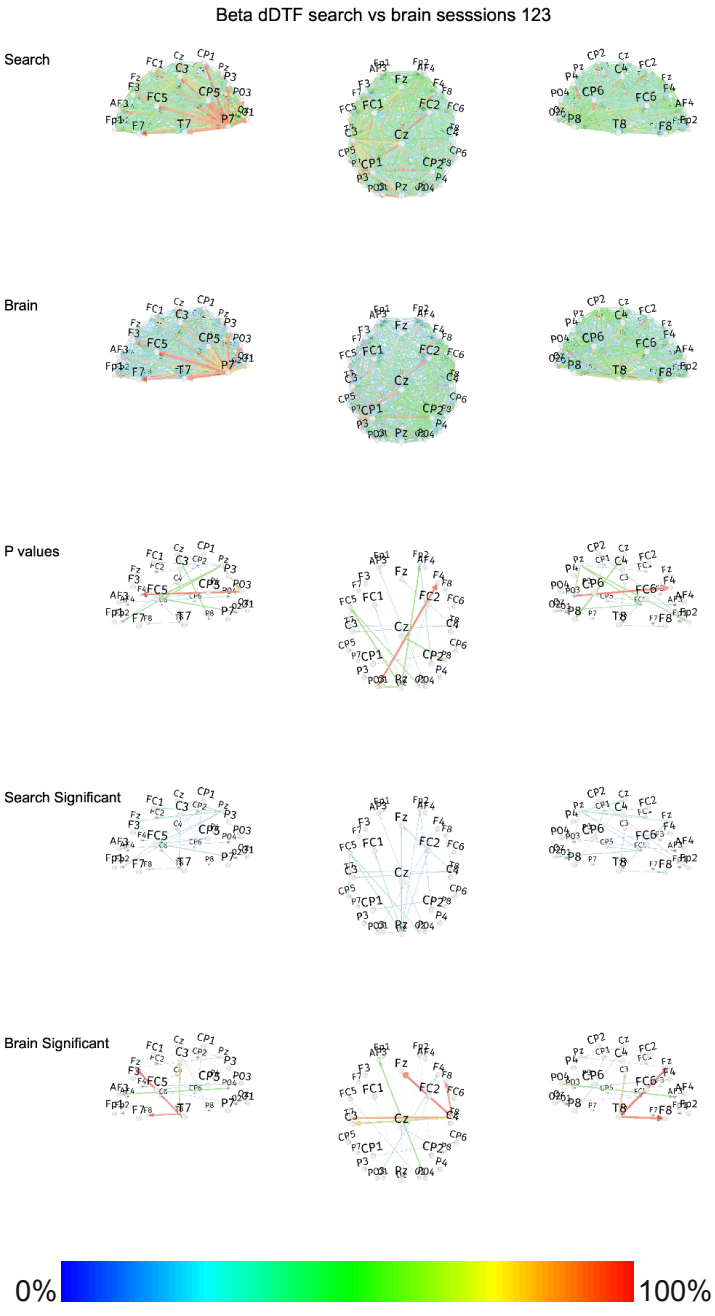


Figure 62. Dynamic Direct Transfer Function (dDTF) for Beta band between Search Engine and Brain-only groups, only for sessions 1,2,3, excluding session 4. Rows 1 (Search Engine group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Theta Band Connectivity

Theta band differences between the groups were pronounced. Brain-only group showed higher theta connectivity (total significant dDTF sum 0.644 vs. 0.331 in Search Engine group). Moreover, 22 connections had larger theta influence in Brain-only group, versus only 4 in Search Engine group, a pattern overwhelmingly favoring the no-tools condition (Figure 63, Appendix AI). This implies that the Brain-only group engaged far more extensive theta band networking, a hallmark of deep cognitive engagement in memory and integrative tasks.

Brain-only group's theta network was characterized by strong fronto-parietal coupling into the right frontal (F4) region. F4 received 11 significant theta connections in Brain-only group (incoming theta sum 0.336, compared to 0.127 in Search Engine group), making it a clear hub for theta band influence. These incoming links originated from widespread sites including left frontal (F3), right parietal (P4), occipital (Oz), and others. Another node, right fronto-central FC2, also saw greater theta input in Brain-only group (5 connections, 0.120 vs. 0.047), further highlighting enhanced fronto-central integration. By contrast, Search Engine group had only minor theta hubs: for instance, frontopolar Fp2 showed slightly higher input in Search Engine group (2 connections; 0.048 vs. 0.017), and midline Cz had a weak Search Engine group advantage (1 link, 0.020 vs. 0.008). These few instances suggest Search Engine group's theta activity was relatively localized (e.g. confined to frontal pole or midline) and with much less networking than Brain-only group's.

All listed theta connections met significance $p < 0.05$; many were in the p ~0.01-0.03 range.

Theta oscillations are known to mediate long-range communication in the brain during complex cognitive operations, such as working memory encoding, retrieval, and integration of information across regions [77]. Our results align with the prior literature: Brain-only group engaged robust fronto-parietal theta connectivity, consistent with greater reliance on internal working memory and executive control to plan and compose the essay. For example, the strong theta inputs to F4 (right frontal cortex) in the Brain-only group most likely demonstrate a coordinated flow of information from posterior areas to a frontal executive node, a pattern often seen when the brain is integrating stored knowledge and monitoring content generation [109,110]. This is in line with research showing that individuals with higher creativity or memory demands exhibit increased fronto-occipital theta coherence [111, 112], reflecting the coupling of visual/semantic regions with frontal planning areas.

In contrast, Search Engine group's much weaker theta connectivity implies that the availability of internet attenuated the need for such intense internal coordination. The internet group could externally offload some memory demands (searching for facts instead of recalling them), which likely reduced frontal theta engagement, indeed, frontal midline theta is an established marker of working memory load and internal focus [77]. Our findings dovetail with the idea that reliance on the internet can redistribute cognitive load [37]: Search Engine group's brains did not have to synchronize distant regions to the same extent, possibly because attention was directed outward (browsing information) rather than inward (retrieving and linking ideas).

Additionally, theta band activity is often linked to sustained attention and episodic memory retrieval [77]. The Brain-only group's stronger theta network may indicate more continuous, self-directed attention to the writing task at hand (since they could not turn to an external source for quick answers), whereas the internet group's attention might have been periodically captured by Search Engine results (potentially engaging different neural processes).

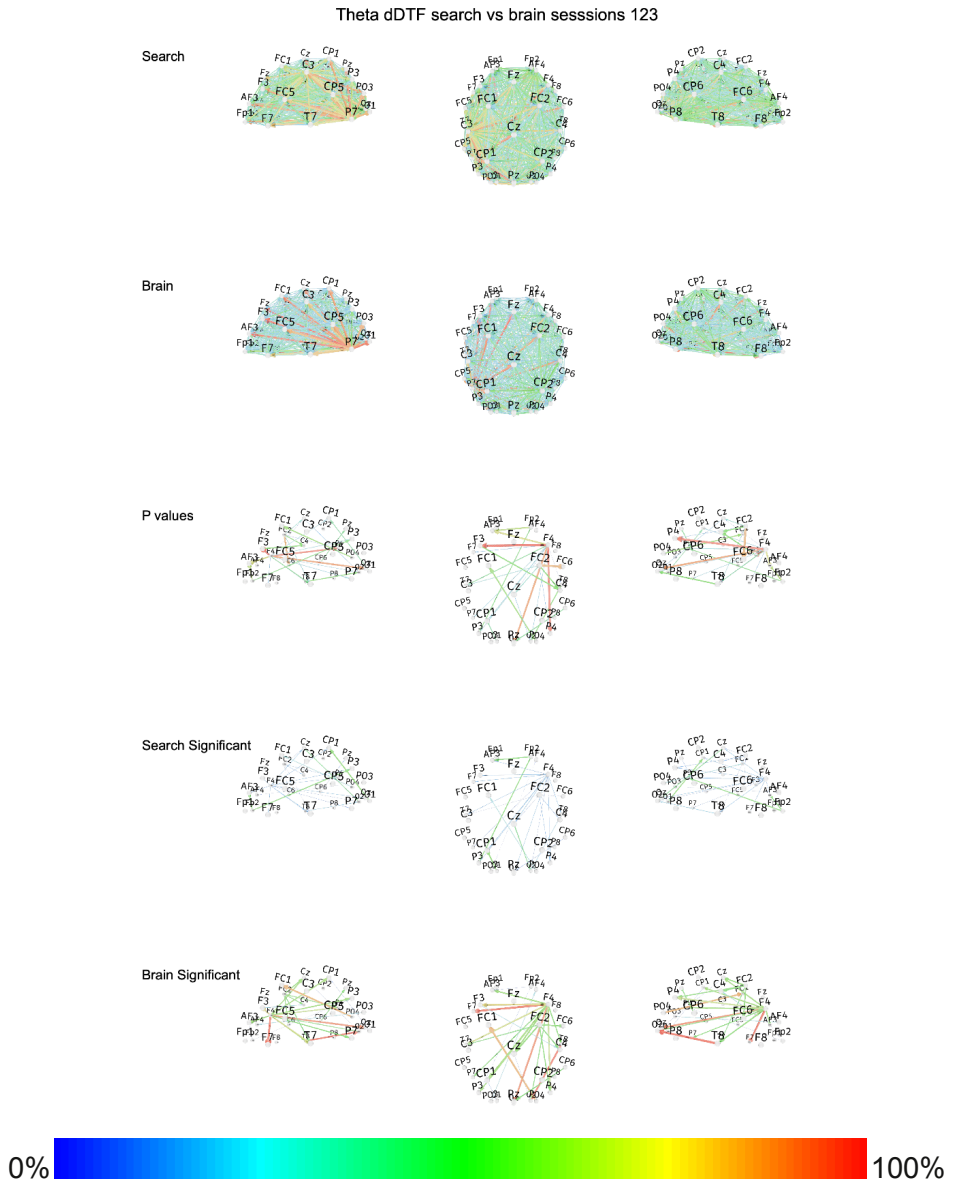


Figure 63. Dynamic Direct Transfer Function (dDTF) for Theta band between Search Engine and Brain-only groups, only for sessions 1,2,3, excluding session 4. Rows 1 (Search Engine group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Delta Band Connectivity

Delta band connectivity showed the largest disparity between the groups. Brain-only group's delta network was far more developed, with the total significant dDTF sum more than double that of the Search Engine group (0.588 vs 0.264). In terms of directionality, 21 delta connections had greater influence in the Brain-only group, vs. only 1 for the Search Engine group (Figure 64, Appendix AB, AE, AH). This pattern indicates that almost all significant delta band interactions were stronger when no external tools were used.

Brain-only group demonstrated widespread delta influences coming from and converging on multiple regions. Notably, several bilateral regions acted as strong delta sources in the Brain-only group, for example, P8 and F7 electrodes each sent out 3 significant connections with much higher dDTF values in the Brain-only group. Brain-only group also had unique delta outflows from areas like O2 (right occipital) and F3 (left frontal), which were minimal in the Search Engine group.

On the receiving end, the Brain-only group's brain had far stronger delta inputs to right-hemisphere regions. For instance, right temporal T8 was a major delta sink with 4 incoming links in the Brain-only group (total 0.204 vs just 0.044 in the Search Engine group). Likewise, right frontal F8 and right frontal F4 each received 3 delta connections in the Brain-only group (sums ~0.07-0.08) compared to ~0.02-0.03 in the Search Engine group. Brain-only group engaged a diffuse network of slow wave interactions linking frontal, temporal, and parietal nodes (predominantly in the right hemisphere). Overall, Search Engine group's delta activity was minimal and lacked the rich coupling seen in Brain-only group.

When examining low vs. high-delta sub-bands (Figure 65), the dominance of the Brain-only group remained evident. In the low-frequency delta range, the Brain-only group's total connectivity was 1.051 vs. 0.537 in the Search Engine group. The Brain-only group had 32 low-delta connections stronger (versus 6 for the Search Engine group). This band showed Brain-only group heavily networking regions like T8 and F8, T8 received 9 low-delta links (sum 0.472) and F8 received 8 links (0.187) in Brain-only group. High-delta had a similar pattern: Brain-only group sum 0.637 vs. 0.261 (Search Engine group), with 26 connections favoring Brain-only group vs. only 1 for Search Engine group. High-delta again highlighted Brain-only group's fronto-temporal and fronto-parietal links were among top connections (both significantly larger in Brain-only group). These sub-band results reinforce that the Brain-only group engaged slow cortical oscillations broadly, whereas the Search Engine group's brain showed weaker delta interactions.

Many of these delta differences were not only statistically significant but highly significant. For example, the F7→CP6 connection in high-delta had $p \approx 0.002$, and F4→F3 in low-delta (directed from right frontal to left frontal) had $p \approx 0.0009$ indicating very strong evidence of group differences. All considered connections had $p < 0.05$ by the analysis design.

Delta band activity in cognitive tasks is less studied, but it has been implicated in attention, motivational processes, and the coordination of large-scale brain networks, especially under

high cognitive demand or fatigue [82]. The significantly elevated delta connectivity in Brain-only group may reflect the brain's recruitment of broad, low-frequency networks to synchronize distant regions when engaging in an effortful internal task (formulating an essay from memory). Such slow oscillatory coupling could underlie the internally directed attention state in the Brain-only group. In essence, without an external knowledge source, participants might tap into a default-mode or memory-related network that operates on delta/theta timescales, integrating emotional, memory, and self-referential processes relevant to creative writing. This is supported by creativity research showing that internally-driven idea generation can involve increased low-frequency coherence across frontal and temporal areas [81].

Overall, the lack of significant delta connectivity in Search Engine group aligns with a more externally oriented cognitive mode: their focus on screen information could engage faster oscillations (alpha/beta for visual-motor processing) and reduce the need for slow, integrative rhythms. Additionally, the aforementioned literature on internet use suggests that having information available instantly can reduce the depth of internal processing (sometimes also described as more shallow or rapid cognitive probing) [37]. Our findings of diminished slow-band coherence in the internet group are consistent with this idea, their brains might not enter the same deep integrative state as those working from memory in the Brain-only group.

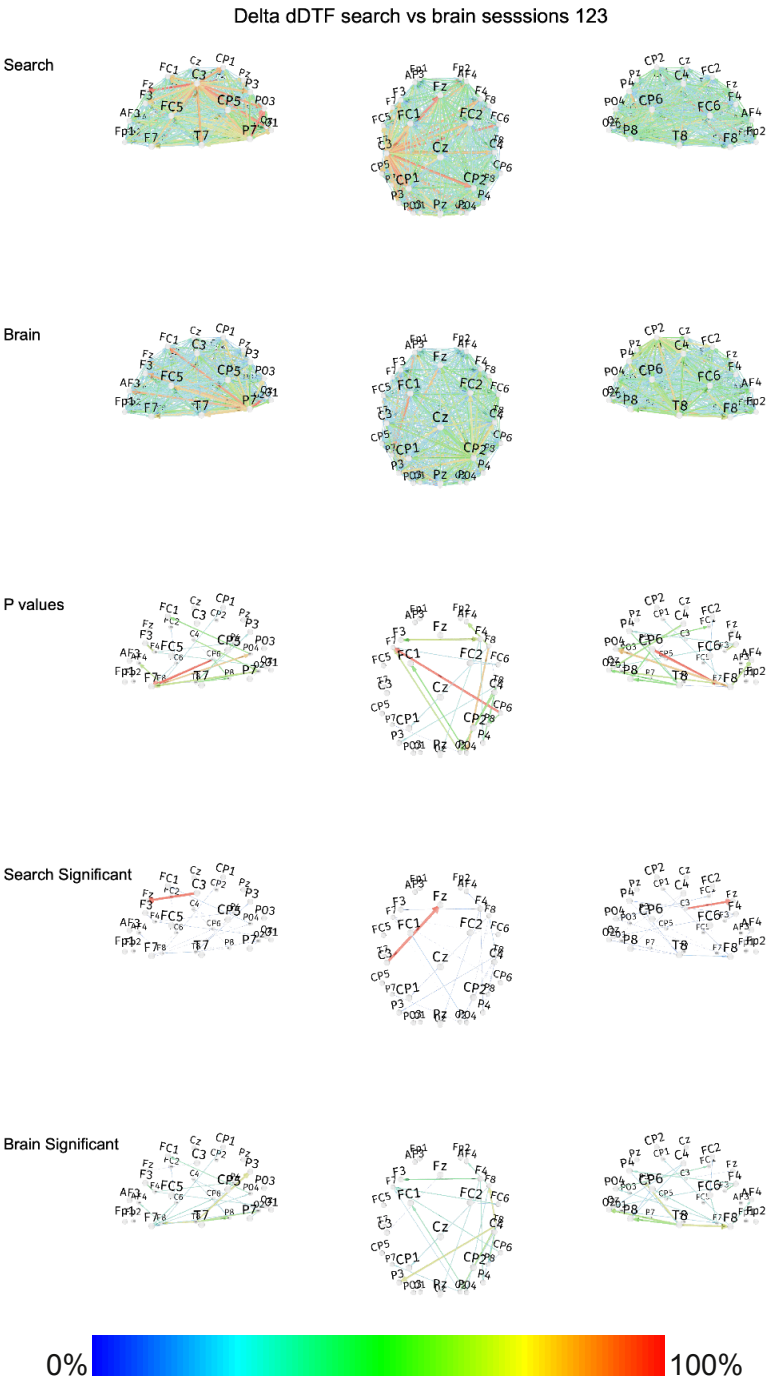


Figure 64. Dynamic Direct Transfer Function (dDTF) for Delta band between Search Engine and Brain-only groups, only for sessions 1,2,3, excluding session 4. Rows 1 (Search Engine group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Summary

Across all frequency bands, **the Brain-only group** demonstrated a **more extensive and stronger connectivity network** during the essay writing task than the **Search Engine group**. This divergence was especially notable in the **lower-frequency bands (delta and theta)**, which are commonly associated with **internalized cognitive processes such as episodic memory retrieval, conceptual integration, and internally focused attention** [77]. In the Brain-only group, the delta-theta range facilitated robust fronto-temporal and fronto-parietal communication, with numerous significant influences converging on frontal executive regions (e.g. F4) from parietal-occipital sources. Such patterns suggest that **without internet assistance, participants engaged memory and planning networks intensely, aligning with the need to recall information and creatively generate content**. This assumption is supported by literature where increased fronto-parietal and fronto-occipital theta coherence is linked to higher creativity and working memory load [81].

Search Engine group, on the other hand, while still performing the complex task of writing, displayed a different connectivity signature. With the ability to search for support online, these participants likely offloaded some cognitive demands, for instance, instead of remembering facts, they could find them, and instead of internally cross-referencing knowledge, they could verify those via web sources. Our results show that this translated to lower engagement of slow integrative rhythms and a shift toward certain higher-frequency connections. The Search Engine group had relatively greater beta-band connectivity to midline parietal regions (Pz). These differences resonate with the notion that internet use can alter cognitive mechanisms [37].

The **cognitive load** also seemed to be managed differently: rather than internally networking brain regions (as Brain-only group did), Search Engine group's strategy leaned on rapid access to information, which might involve more localized or task-specific circuits. For example, the prominent Pz hub in Search Engine group's beta network could indicate focal integration of visual input and top-down attention on the external content, consistent with prior research that beta oscillations support maintaining attention on currently processed stimuli [80].

In summary, the Brain-only group's connectivity suggests a state of **increased internal coordination**, engaging memory and creative thinking (manifested as theta and delta coherence across cortical regions). The Engine group, while still cognitively active, showed a tendency toward **more focal connectivity** associated with handling external information (e.g. beta band links to visual-parietal areas) and comparatively **less activation of the brain's long-range memory circuits**. These findings are in line with literature: tasks requiring internal memory amplify low-frequency brain synchrony in frontoparietal networks [77], whereas outsourcing information (via internet search) can reduce the load on these networks and alter attentional dynamics. Notably, prior studies have found that practicing internet search can reduce activation in memory-related brain areas [83], which dovetails with our observation of weaker connectivity in those regions for Search Engine group. Conversely, the richer connectivity of Brain-only group may reflect a cognitive state akin to that of high performers in creative or memory tasks, for instance, high creativity has been associated with increased

fronto-occipital theta connectivity and intra-hemispheric synchronization in frontal-temporal circuits [81], patterns we see echoed in the Brain-only condition.

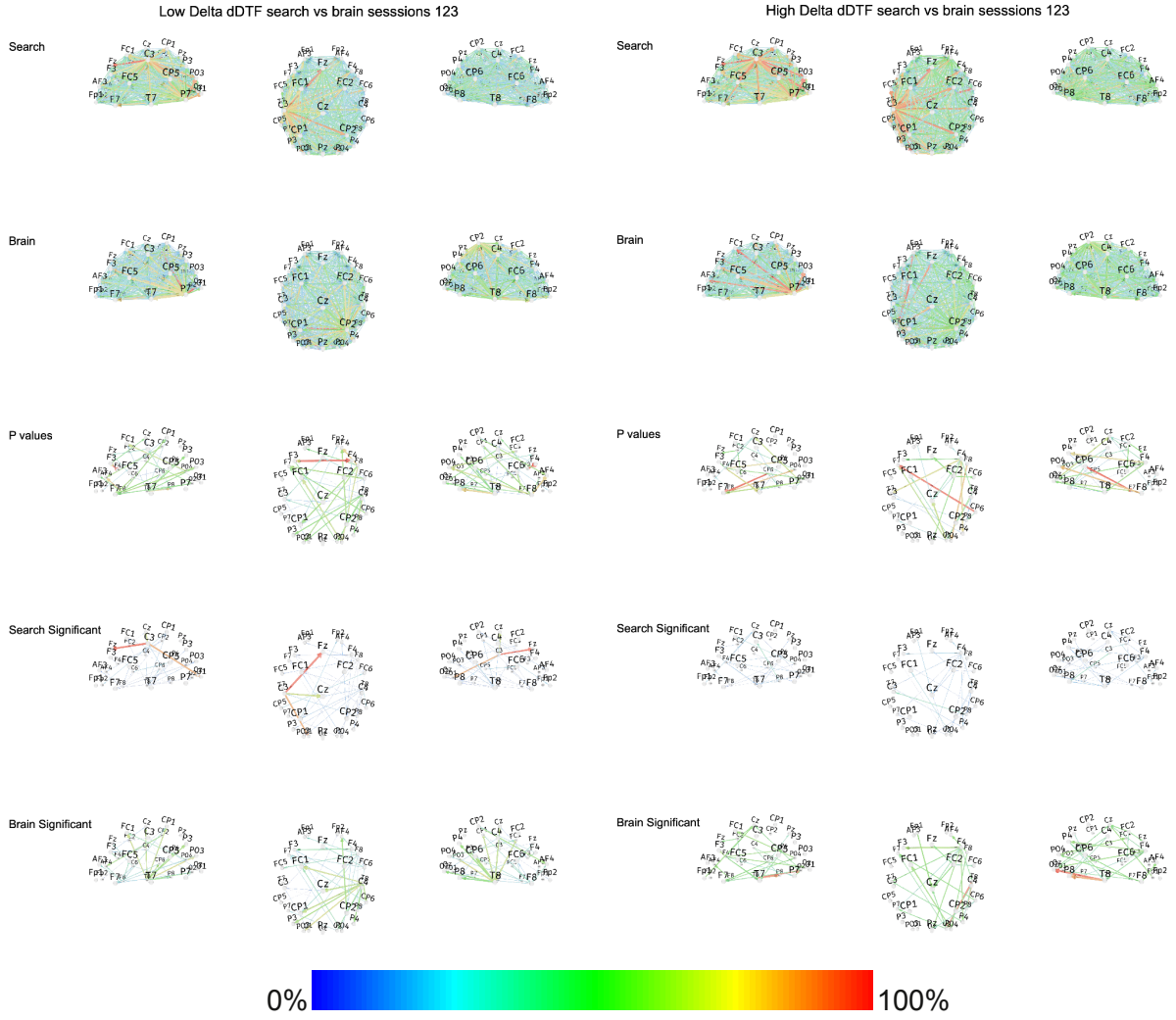


Figure 65. Dynamic Direct Transfer Function (dDTF) for Low Delta and High Delta bands between Search Engine and Brain-only groups, only for sessions 1,2,3, excluding session 4. Rows 1 (Search Engine group) and 2 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

EEG Results: LLM Group vs Search Engine Group

Alpha Band Connectivity

In the alpha range (8-12 Hz), both groups demonstrated comparable overall dDTF strength, with the Search Engine group slightly exceeding the LLM group (0.901 vs. 0.891). However, the directionality and network topology diverged (Figure 66-67, Appendix S, P, V). Search Engine group exhibited significantly elevated parieto-frontal inflow targeting the AF3 region, a prefrontal electrode associated with attentional control and inhibition, particularly from occipital and parietal regions (P7, PO3, Oz). Low-alpha analysis reinforced this trend. The Search Engine group again exhibited greater AF3-directed inflow, particularly from posterior hubs. High-alpha activity, in contrast, was marginally higher in the LLM group.

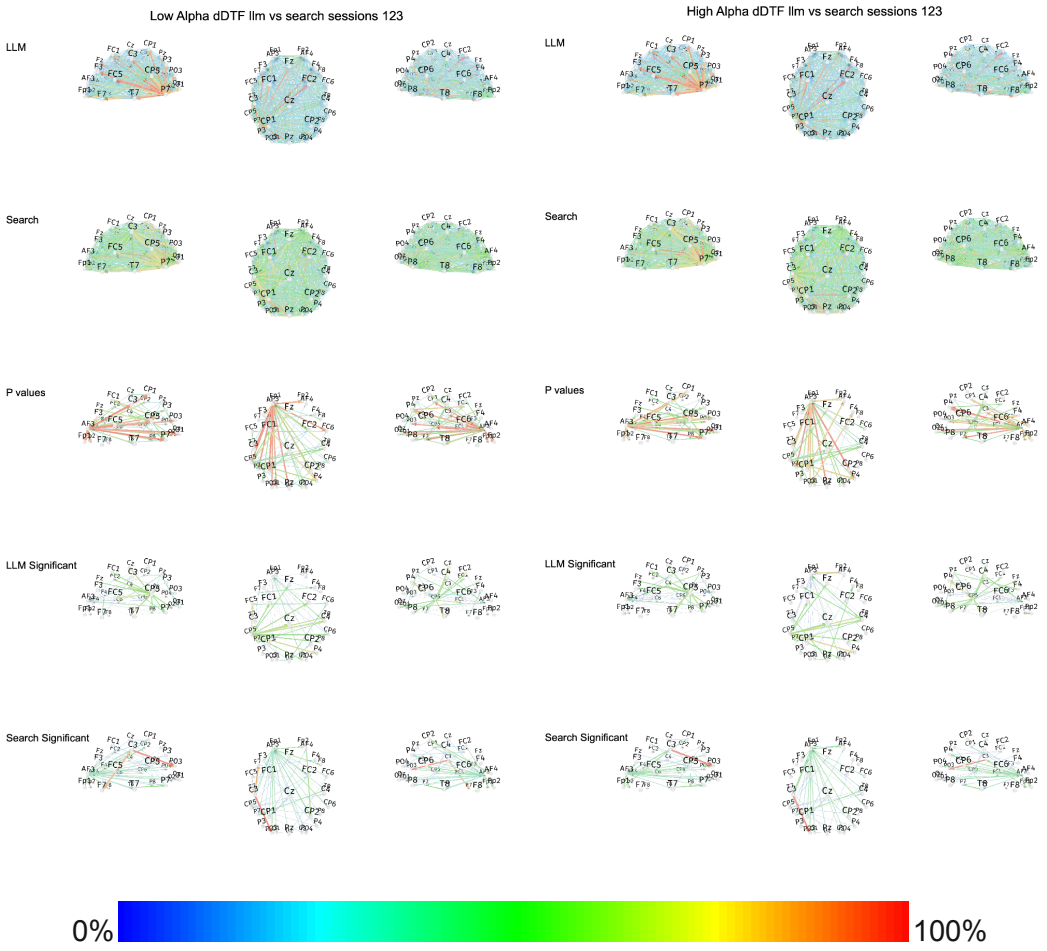


Figure 66. Dynamic Direct Transfer Function (dDTF) for Low Alpha, Alpha, High Alpha bands between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

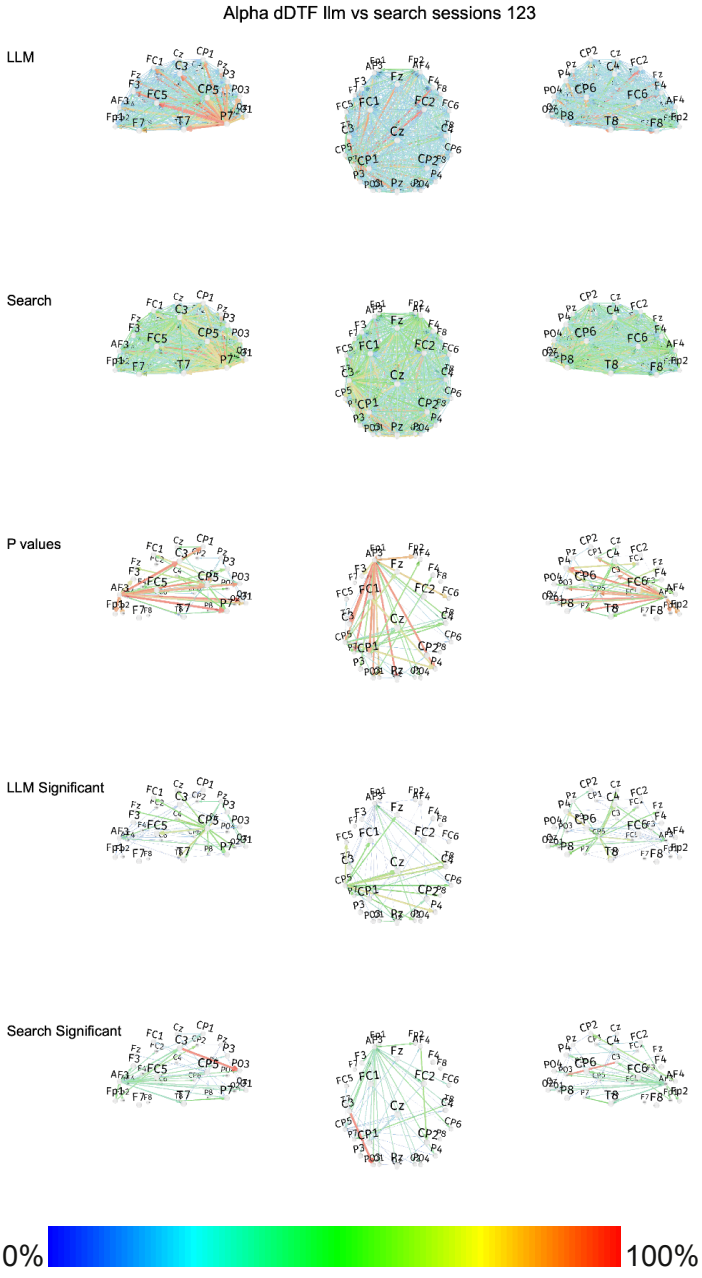


Figure 67. Dynamic Direct Transfer Function (dDTF) for Alpha between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Beta Band Connectivity

Beta band findings point to sharply contrasting motor and executive network activations. The LLM group consistently demonstrated stronger outflow from motor-associated regions (e.g. CP5, FC6), especially in the high-beta range (13-30 Hz). These connections likely represent procedural fluency and feedback loops tied to text generation via typing and interaction with an LLM.

In low-beta frequencies (Figure 68-69, Appendix T, Q, W), Search Engine group displayed enhanced directed flow toward AF3 from posterior and parietal sources, indicating more top-down control over cognitive Search Engine processes. The balance of connectivity suggests that while LLM group offloaded cognitive load to an LLM, Search Engine group recruited more endogenous executive regulation to curate and synthesize information from online sources.

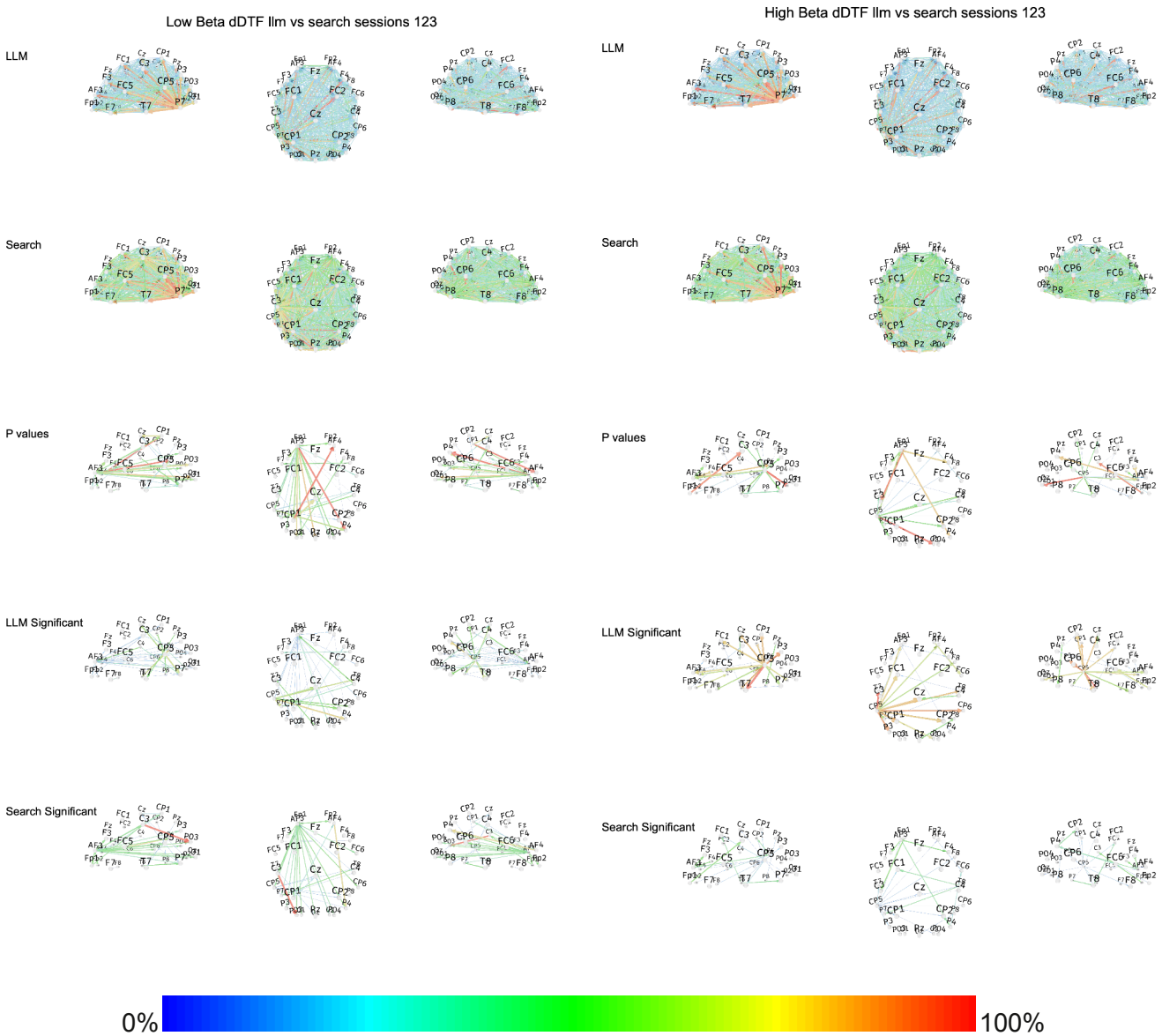


Figure 68. Dynamic Direct Transfer Function (dDTF) for Low Beta, High Beta bands between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

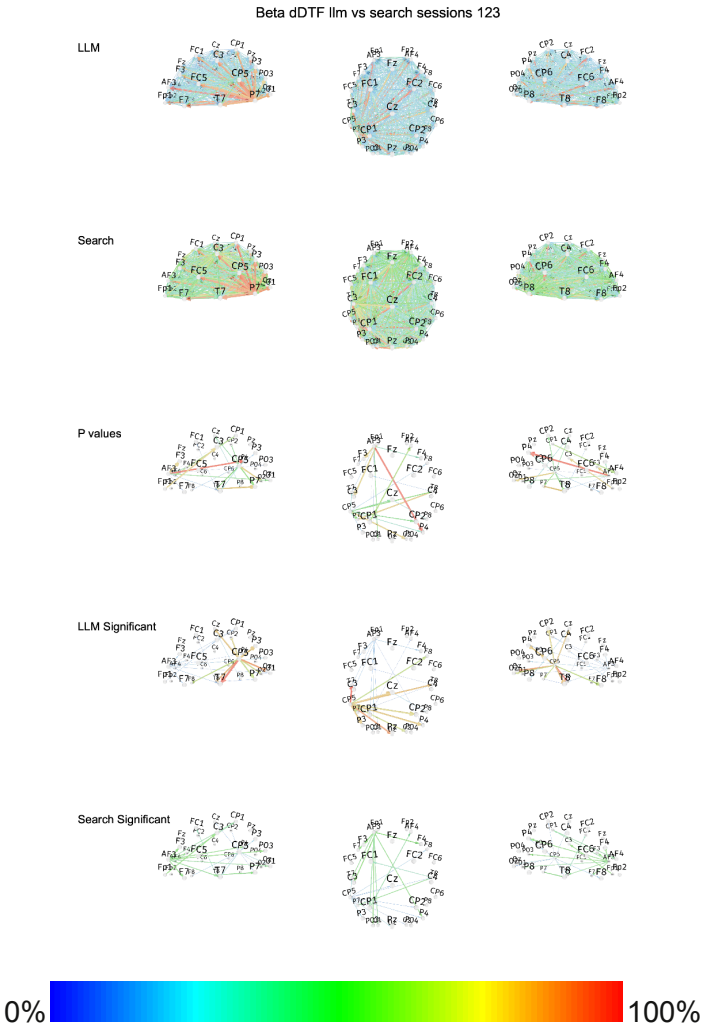


Figure 69. Dynamic Direct Transfer Function (dDTF) for Beta band between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Theta Band Connectivity

Theta band activity revealed stronger global connectivity for the LLM group (0.920 vs. 0.826). This was particularly evident in connections from parietal (P7) and central (CP5) regions toward frontal targets like AF3 (Figure 70, Appendix Y). Theta oscillations are linked to working memory and semantic processing [84].

Despite the overall lower dDTF magnitude, Search Engine group exhibited more posterior-to-frontal connections into AF3, including from PO3 and C3, reinforcing the hypothesis that Search Engine users relied more on visual-spatial memory.

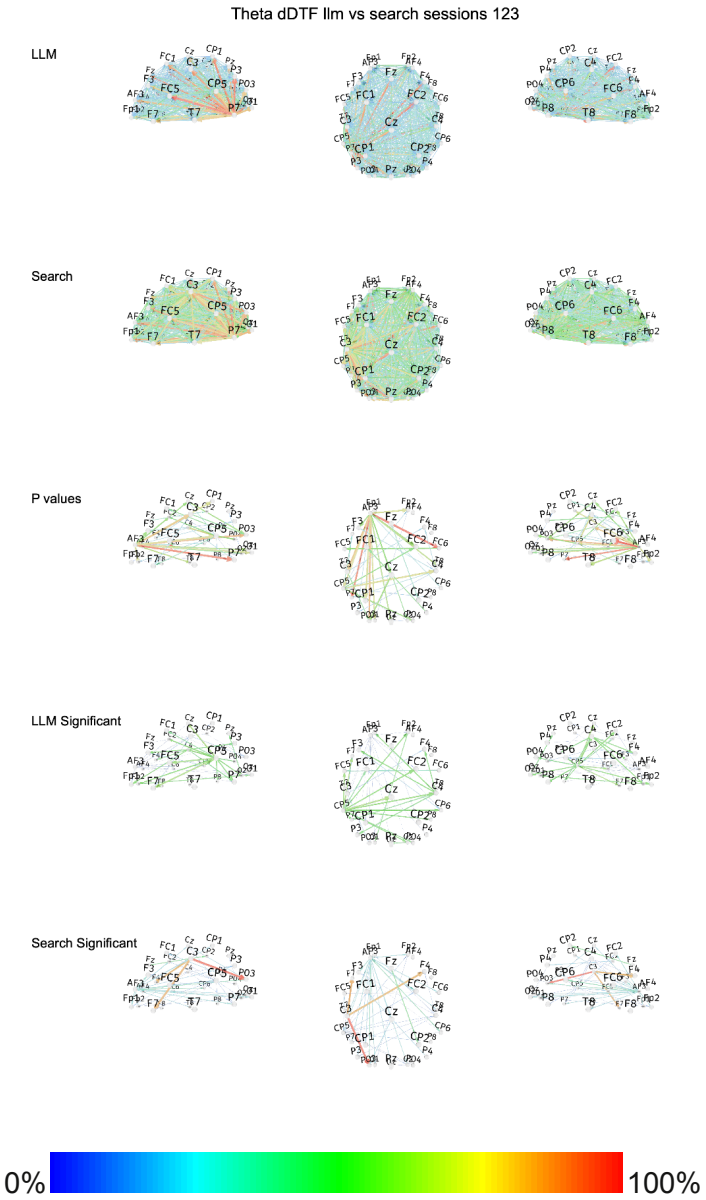


Figure 70. Dynamic Direct Transfer Function (dDTF) for Theta band between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all

pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Delta Band Connectivity

The LLM group showed greater total connectivity in high-delta, whereas the Search Engine group led in low-delta bands (Figure 71-72, Appendix R, X, U). The delta band, typically linked with homeostatic and motivational processes [85], reflected deeper frontal-subcortical control engagement in the Search Engine group, with strong and significant AF3 inflows from posterior regions including CP6 and O2.

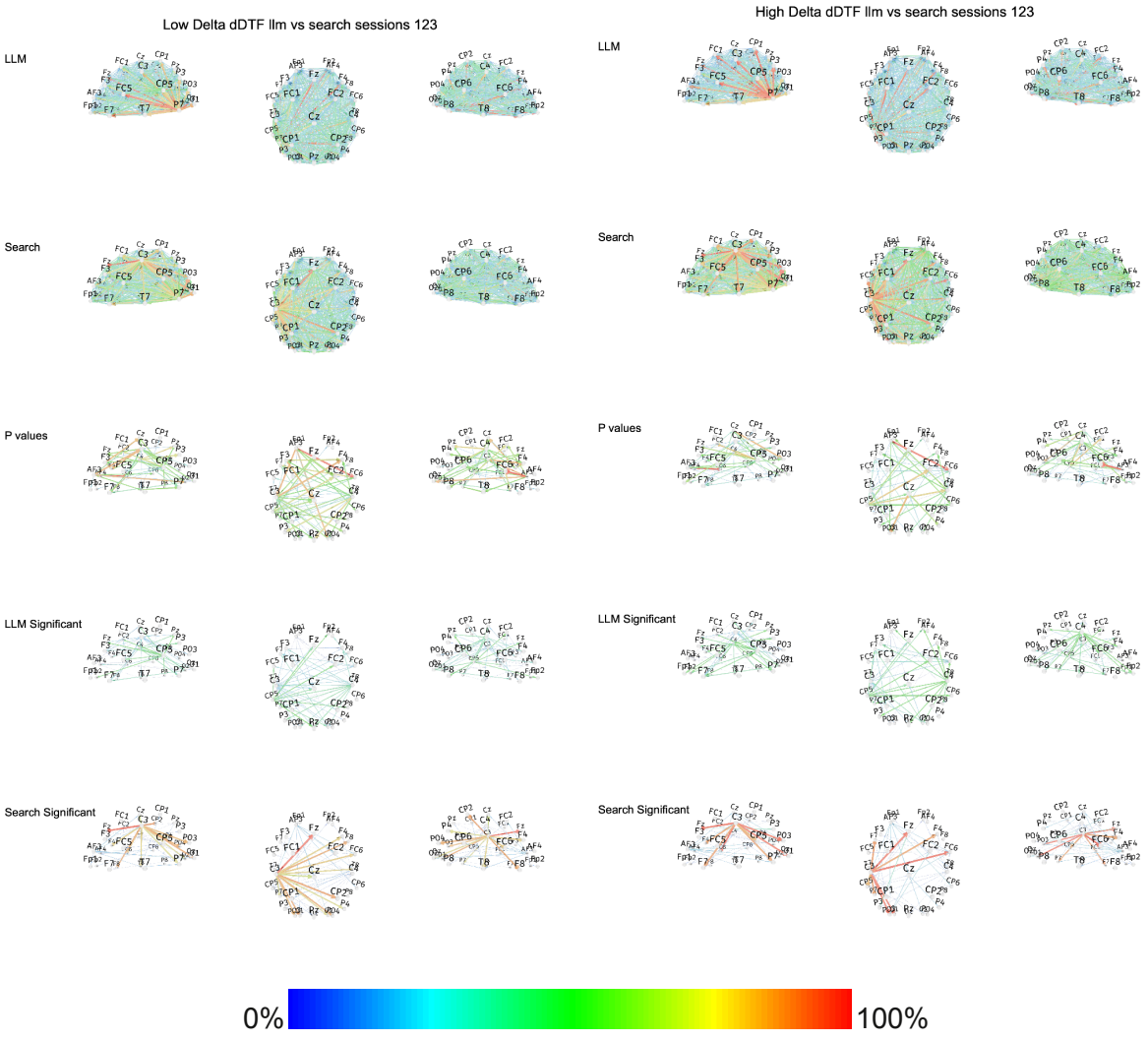


Figure 71. Dynamic Direct Transfer Function (dDTF) for Low Delta, High Delta bands between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least

significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

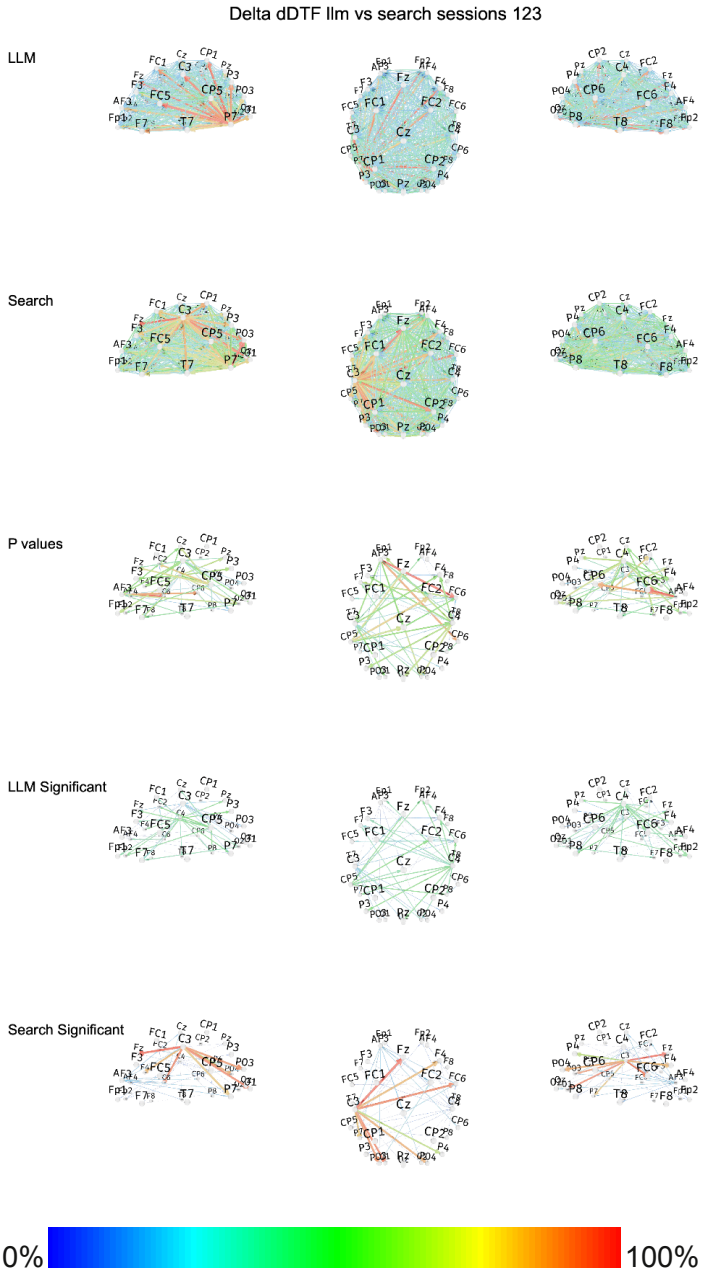


Figure 72. Dynamic Direct Transfer Function (dDTF) for Delta band between LLM and Search Engine groups, only for sessions 1,2,3, excluding session 4. Rows 1 (LLM group) and 2 (Search Engine group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Third row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last two rows show only significant dDTF values filtered using the third row of p values, and

normalized by the min and max ones in rows 4 and 5. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Summary

Using AI writing tools vs. internet Search Engine engages different neurocognitive dynamics: Search Engine group showed connectivity patterns consistent with higher external information load, engaging memory retrieval and visual-executive integration (especially in alpha/theta bands), while LLM group exhibited greater internal executive network coherence and bilateral integration (especially in beta/delta bands), consistent with planning, and potentially more efficient cognitive processing.

These results suggest that AI assistance in writing may free up cognitive resources (reducing memory load) and allow the brain to reallocate effort toward executive functions, whereas traditional Search Engine-based writing engages the brain's integrative and memory systems more strongly. This dichotomy reflects two distinct cognitive modes: externally scaffolded automation versus internally managed curation. The directionality of dDTF differences underscores how cognitive workflows differ: the Search Engine group's brain network was more bottom-up, and the tool group's more top-down, mirroring their distinct approaches to essay composition.

Session 4

Brain

Band	**Most Frequent Sessions Pattern**	**Count**	**Significance**
Alpha	2 > 3 > 4 > 1	7	**, *
Beta	3 > 4 > 2 > 1	10	**, *
Delta	2 > 3 > 4 > 1	6	***, **, *
High Alpha	2 > 3 > 4 > 1	8	**, *
High Beta	3 > 2 > 4 > 1	6	**, *
High Delta	2 > 3 > 4 > 1	8	**, *
Low Alpha	2 > 4 > 3 > 1	7	**, *
Low Beta	3 > 2 > 4 > 1	8	**, *
Low Delta	2 > 4 > 3 > 1	5	**, *
Theta	2 > 4 > 3 > 1	12	**, *

Table 2. Summary of dDTF differences across Brain-only sessions for each EEG frequency band. Boldface indicates the session with the highest connectivity in that band. Arrows denote relative ordering of connectivity strength. Significance marked with asterisks as following: Highly significant (***), Strong evidence (**), Moderate evidence (*). For detailed summary check Appendix AJ-AS.

As a reminder, the fourth session of our study was executed in a different manner from Sessions 1, 2, 3.

During Session 4, participants were reassigned to the group opposite of their original assignment from Sessions 1, 2, 3. Session 4 was not a mandatory session, and thus, due to participants' availability and scheduling constraints, only 18 participants were able to attend. These individuals were placed in either LLM group or Brain-only group based on their original group placement (e.g. participant 17, originally assigned to LLM group for Sessions 1, 2, 3, was reassigned to Brain-only group for Session 4). Thus, we refer to all participants who originally performed Sessions 1, 2, 3 as Brain-only group, **Brain-to-LLM group** for Session 4, as they performed their 4th session as LLM group. As for participants who originally performed Sessions 1, 2, 3 as an LLM group, we refer to them as the **LLM-to-Brain group** for Session 4, as they performed their 4th session as a Brain-only group.

Additionally, instead of offering a new set of three essay prompts for session 4, we offered participants a set of personalized prompts made out of the topics **each** participant already wrote about in sessions 1, 2, 3. For example, participant 17 picked up Prompt CHOICES in session 1, Prompt PHILANTHROPY in session 2 and prompt PERFECT in session 3, thus getting a selection of prompts CHOICES, PHILANTHROPY and PERFECT to select from for their session 4. The participant picked up CHOICES in this case. This personalization took place for **each** participant who came for session 4.

The participants were not informed beforehand about the reassignment of the groups/essay prompts in session 4.

Thus, in the remainder of this section of the paper, as in any other section of this paper describing Session 4, we only present results for these **18 participants who took part in all 4 sessions**.

Interpretation

Cognitive Adaptation

Here we report how brain connectivity evolved over four sessions of an essay writing task in Sessions 1, 2, 3 for the Brain-only group and Session 4 for the LLM-to-Brain group. The results revealed clear frequency-specific patterns of change: lower-frequency bands (delta, theta, alpha) all showed a dramatic increase in connectivity from the first to second session, followed by either a plateau or decline in subsequent sessions, whereas the beta band showed a more linear increase peaking at the third session. These patterns likely reflect the cognitive adaptation and learning that occurred with repeated writing in our study. Session 1 (first time doing the task) was associated with minimal connectivity across all bands, a plausible indication that novice users had less coordinated brain network engagement, possibly due to uncertainty or the novelty of the task: the participants did not know any details of the study, like the task, the duration, etc. By Session 2, we observed robust increases in connectivity in all bands, suggesting that once participants became familiar with the task and attempted to improve on their essay, having learned about the task duration and other details of the study, their brains recruited multiple networks more strongly. This aligns with the idea that practice engages memory and control processes more deeply: for instance, the large rise in theta and alpha

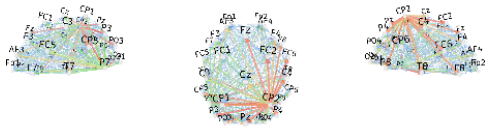
connectivity from Session 1 to 2 is in line with enhanced retrieval of ideas and top-down organization in the second writing session [86]. The delta band's significant spike at Session 2 may indicate a surge in focused attention as participants refined their work (Figure 73, Appendix AJ-AS) [85]. By Session 3, some of these networks (alpha, theta, delta) showed decline, which could be attributed to diminishing returns of practice or mental fatigue. Interestingly, beta band connectivity continued to rise into Session 3, which might reflect that certain higher-order processes (like active working memory usage, and fine-grained attention) kept improving with each iteration [87-95]. Beta oscillations support the active maintenance of task information and long-range cortical interactions [89-92]; the Session 3 peak in beta (Figure 75) suggests that by the third session, participants were potentially coordinating distant brain regions (e.g. frontal and occipital) to a greater extent, perhaps as they polished the content and structure of their essays.

The critical point of this discussion is Session 4, where participants wrote without any AI assistance after having previously used an LLM. Our findings show that Session 4's brain connectivity did not simply reset to a novice (Session 1) pattern, but it also did not reach the levels of a fully practiced Session 3 in most aspects. Instead, Session 4 tended to mirror somewhat of an intermediate state of network engagement. For example, in the alpha and beta bands, which are associated with internally driven planning, critical reasoning, and working memory, Session 4 connectivity was significantly lower than the peaks observed in Sessions 2-3 (alpha), Figure 74, or Session 3 (beta), yet remained above the Session 1 which we consider a baseline in this context. One plausible explanation is that the LLM had previously provided suggestions and content, thereby reducing the cognitive load on the participants during those assisted sessions. When those same individuals wrote without AI (Session 4), they may have leaned on whatever they learned or retained from the AI, but because prior sessions did not require the significant engagement of executive control and language-production networks, engagement we observed in Brain-only group (see Section 'EEG Results: LLM Group vs Brain-only Group' for more details), the subsequent writing task elicited a reduced neural recruitment for content planning and generation.

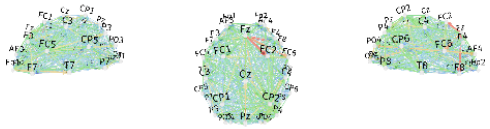
Cognitive offloading to AI

This interpretation is supported by reports on **cognitive offloading to AI**: reliance on AI systems can lead to a passive approach and diminished activation of critical thinking skills when the person later performs tasks alone [3]. In our context, the lower alpha connectivity in Session 4 (relative to Sessions 2-3) could indicate less activation of top-down executive processes (such as internally guided idea generation), consistent with the notion that the LLM had taken some of that burden earlier, leaving the participants with weaker engagement of those networks. Likewise, the drop in beta band coupling in Session 4 suggests a reduction in sustained working memory usage compared to highly practiced (Session 3) participants [88]. This resonates with findings that frequent AI tool users often bypass deeper engagement with material, leading to 'skill atrophy' in tasks like brainstorming and problem-solving [96]. In short, Session 4 participants might not have been leveraging their full cognitive capacity for analytical and generative aspects of writing, potentially because they had grown accustomed to AI support.

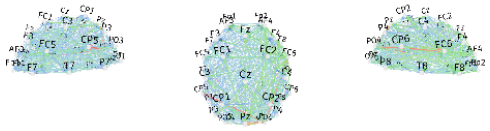
Sessions 1



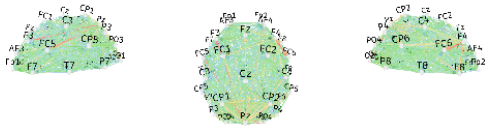
Sessions 2



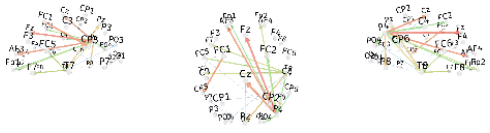
Sessions 3



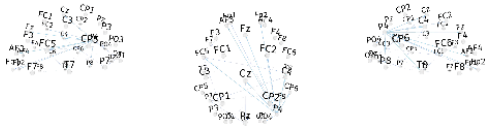
Sessions 4



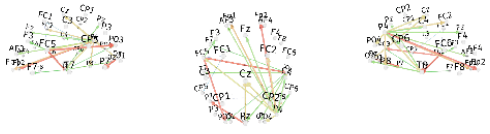
P values



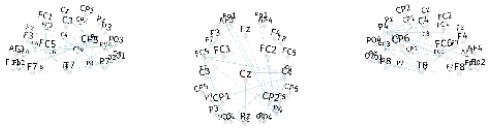
Sessions 1
Significant



Sessions 2
Significant



Sessions 3
Significant



Sessions 4
Significant



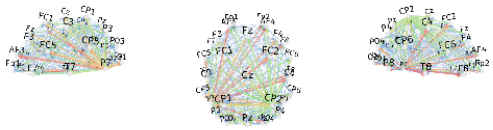


Figure 73. Dynamic Direct Transfer Function (dDTF) for Delta band for Brain-only group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

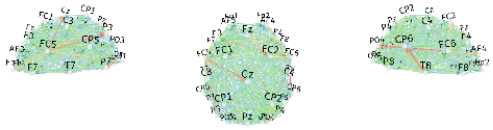
Cognitive processing

On the other hand, Session 4's connectivity was not universally down, in certain bands, it remained relatively high and even comparable to Session 3. Notably, theta band connectivity in Session 4, while lower in total than Session 3, showed several specific connections where Session 4 was equal or exceeded Session 3 (e.g. many connections followed S2 > S4 > S3 > S1 pattern). Theta is often linked to semantic retrieval and creative ideation; the maintained theta interactions in Session 4 may reflect that these participants were still actively retrieving knowledge or ideas, possibly recalling content that AI had provided earlier. This might manifest as, for example, remembering an outline or argument the AI suggested and using it in Session 4, as several participants reported during the interview phase. In a sense, the AI could have served as a learning aid, providing new information that the participants internalized and later accessed. The data hints at this: one major theta hub in all sessions was the frontocentral area FC5 (near premotor/cingulate regions), involved in language and executive function, which continued to receive strong inputs in Session 4. Therefore, even after AI exposure, participants engaged brain circuits for memory and planning. Similarly, the delta band in Session 4 remained as active as in Session 3, indicating that sustained attention and effort were present. This finding is somewhat encouraging: it suggests that having used AI did not make the participants completely disengaged or inattentive when they later wrote on their own. They were still concentrating, delta connectivity at Session 4 was ~45% higher than Session 1's and matched Session 3's level. One interpretation is that the challenge of writing without assistance, after being used to it, may have demanded a refocusing of attention, thereby elevating low-frequency oscillatory coordination similar to a practiced task. In other words, Session 4 required the participants perhaps to compensate for the lack of AI, which aligns with delta oscillations' role in inhibiting external distractions and maintaining task focus [85]. This paints a nuanced picture: prior AI help did not leave participants unengaged when the AI was removed, they still harnessed cognitive effort (as seen in theta/delta activity), but their engagement tilted away from the higher-frequency processes (alpha/beta) that underpin self-driven idea organization and reasoning.

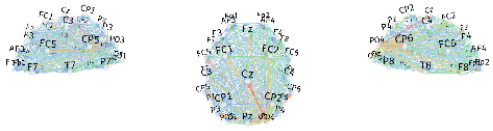
Sessions 1



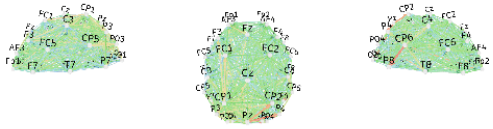
Sessions 2



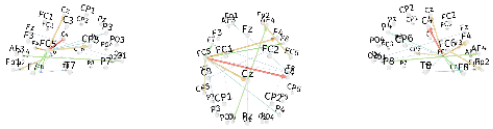
Sessions 3



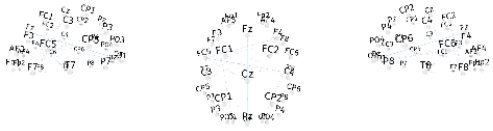
Sessions 4



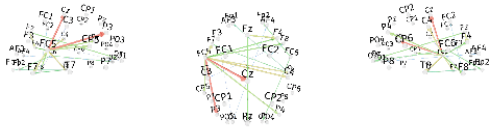
P values



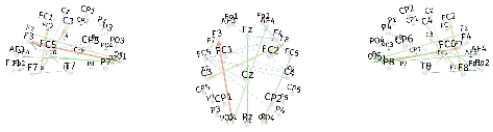
Sessions 1
Significant



Sessions 2
Significant



Sessions 3
Significant



Sessions 4
Significant



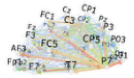


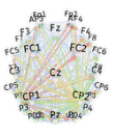
Figure 74. Dynamic Direct Transfer Function (dDTF) for Alpha band for Brain-only group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

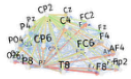
There are important cognitive and educational implications of these findings. The differences between Session 4 and the Brain-only sessions 1, 2, 3 suggest that AI tools can alter the balance of cognitive processes involved in writing. With repeated unassisted practice (Sessions 1, 2, 3), participants progressively strengthened networks associated with planning, language, and attentional control, essentially exercising a broad spectrum of brain regions to improve their essays. In contrast, the Session 4 scenario (having had AI support earlier) seems to limit some of this integration: the participants may have achieved competency in content via AI, but perhaps without engaging fully in the underlying cognitive work. As a result, when writing alone, they showed signs of a *less coordinated neural effort* in most bands. **This could translate behaviorally into writing that is adequate (since most of them did recall their essays as per the interviews) but potentially lacking in originality or critical depth. N-grams analysis supports this claim: as an example, LLM-to-Brain group reused 'before speaking' n-gram, which was actively used by LLM group before in session 2,** (Figure 83, Figure 85)**, topics FORETHOUGHT and PERFECT. Simultaneously, we can see how human teachers scored the essays in these two topics low on the metric of uniqueness** among other metrics (Figure 51, Figure 54)**.** Such an interpretation aligns with concerns that over-reliance on AI can erode critical thinking and problem-solving skills: users might become good at using the tool but not at performing the task independently to the same standard. Our neurophysiological data provides the initial support for this process, showing concrete changes in brain connectivity that mirror that shift.

Our results also caution that certain neural processes require active exercise. The under-engagement of alpha and beta networks in post-AI writing might imply that if a participant skips developing their own organizational strategies (because an AI provided them), those brain circuits might not strengthen as much. Thus, when the participant faces a task alone, they may underperform in those aspects. In line with this, recent research has emphasized the need to balance AI use with activities that build one's own cognitive abilities [3]. From a neuropsychological perspective, our findings underscore a similar message: the brain adapts to how we train it. If AI essentially performs the high-level planning, the brain will allocate less resources to those functions, as seen in the moderated alpha/beta connectivity in Session 4.

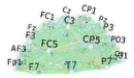
Sessions 1

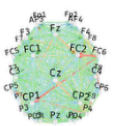


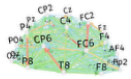




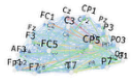
Sessions 2

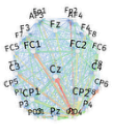


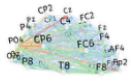




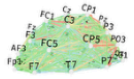
Sessions 3

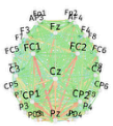


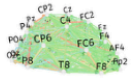




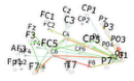
Sessions 4

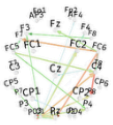


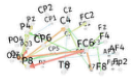




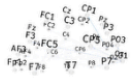
P values



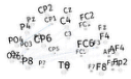




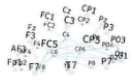
Sessions 1
Significant

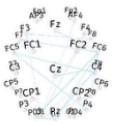






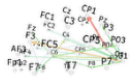
Sessions 2
Significant

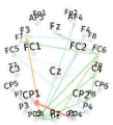


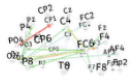




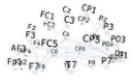
Sessions 3
Significant







Sessions 4
Significant



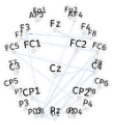




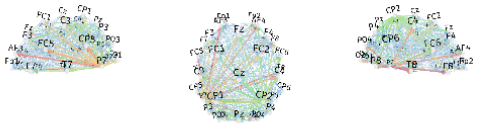


Figure 75. Dynamic Direct Transfer Function (dDTF) for Beta band for Brain-only group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

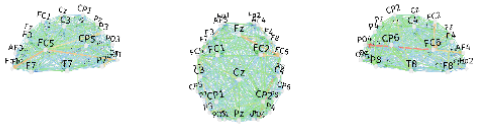
Active learning and practice drove the brain to form stronger networks (as seen in Session 2's across-the-board connectivity surge).

Interestingly, **Session 2 consistently showed the peak in delta, theta, and alpha**, even higher than Session 3 for some bands (Figure 76). This could be due to the nature of the study and task sequence: Session 1 was an initial session, participants did not know anything about the nature of the task, and Session 2 was likely a significant improvement as they knew the task and details about it. By Session 3, however, there might have been diminishing scope for improvement or novelty, resulting in slightly lower engagement (except in beta, possibly due to fine-tuning processes still increasing). Session 4 participants, on the other hand, had a different prior experience: their 'Session 2 and 3' involved help from an LLM. So their Session 4 was effectively the first solo 'revision' of an essay writing task after AI involvement. They demonstrated some increased connectivity (relative to an initial attempt) as discussed, but not the dramatic spike a non-AI user got in their first 'revision' (Session 2). This discrepancy might indicate that AI-assisted revisions do not stimulate the brain as much as tools-free revisions. When the AI was used for support in those middle sessions, the users' brains perhaps did not experience the full challenge, so when they confronted the challenge in Session 4, it felt more like a second-hand effort. This interpretation aligns with educational observations that students who rely on calculators or solution manuals heavily can struggle more when those aids are removed; they have not internalized the problem-solving process, which is reflected in their neural activity (or lack thereof) when they try to solve the problem independently.

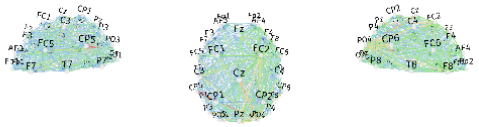
Sessions 1



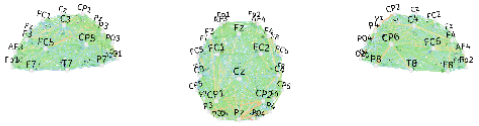
Sessions 2



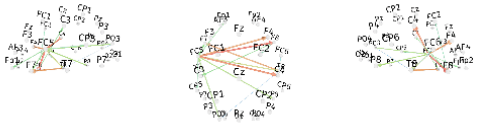
Sessions 3



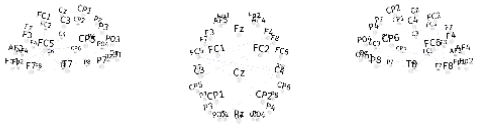
Sessions 4



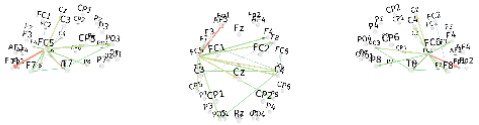
P values



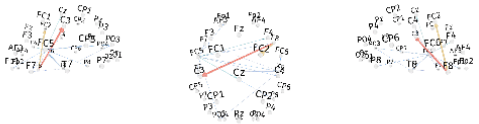
Sessions 1
Significant



Sessions 2
Significant



Sessions 3
Significant



Sessions 4
Significant

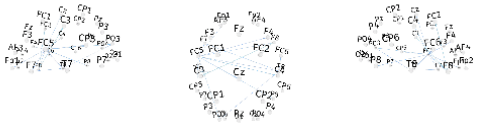




Figure 76. Dynamic Direct Transfer Function (dDTF) for Theta band for Brain-only group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Cognitive 'Deficiency'

In conclusion, our analysis indicates that **repeated essay writing without AI leads to strengthening of brain connectivity in multiple bands, reflecting an increased involvement of memory, language, and executive control networks. Prior use of AI tools, however, appears to modulate this trajectory**. Participants who had AI assistance showed a somewhat reduced connectivity profile in the high-frequency bands when writing on their own, suggesting they might not be engaging in as much self-driven elaboration or critical scrutiny as their counterparts. At the same time, these LLM-to-Brain participants did not entirely disengage, their sustained theta and delta activity pointed to continued cognitive effort, just focused perhaps more on recall than on complex reasoning. These findings resonate with current concerns about AI in education: while AI can be used for support during a task, there may be a **trade-off between immediate convenience and long-term skill development** [96]. Our brain connectivity results provide a window into this trade-off, showing that certain neural pathways (e.g. those for top-down control) may be less engaged when LLM is used. Going forward, a balanced approach is advisable, one that might leverage AI for routine assistance but still challenges individuals to perform core cognitive operations themselves. In doing so, we can harness potential benefits of AI support without impairing the natural development of the brain's writing-related networks.

It would be important to explore hybrid strategies in which AI handles routine aspects of writing composition, while core cognitive processes, idea generation, organization, and critical revision, remain user-driven. During the early learning phases, full neural engagement seems to be essential for developing robust writing networks; by contrast, in later practice phases, selective AI support could reduce extraneous cognitive load and thereby enhance efficiency without undermining those established networks.

LLM

Band	Most Frequent Sessions Pattern	Count	Significance
Alpha	4 > 2 > 1 > 3	11	*
Beta	1 > 4 > 2 > 3	32	*
Delta	4 > 1 > 2 > 3	22	* (*** for some sub-sums)
High Alpha	4 > 2 > 1 > 3	9	*
High Beta	1 > 4 > 2 > 3	23	(** for some sub-sums)
High Delta	4 > 1 > 2 > 3	16	(** for some sub-sums)
Low Alpha	4 > 2 > 1 > 3	7	*
Low Beta	4 > 1 > 2 > 3 4 > 2 > 1 > 3 (tied)	14	*
Low Delta	4 > 1 > 2 > 3	32	* (*** for some sub-sums)
Theta	4 > 2 > 1 > 3	13	*

Table 3. Summary of dDTF differences across LLM sessions for each EEG frequency band. Boldface indicates the session with the highest connectivity in that band. Arrows denote relative ordering of connectivity strength. Significance marked with asterisks as following: Highly significant (***), Strong evidence (**), Moderate evidence (*). For detailed summary check Appendix AT-BC.

Here we report how brain connectivity evolved over four sessions of an essay writing task in Sessions 1, 2, 3 for the LLM group and Session 4 for the Brain-to-LLM group.

Alpha Band: Total dDTF was higher in Session 4 than in Sessions 1, 2, 3. The sum of significant connections in Session 4 was 0.823 (versus 0.547, 0.285, 0.107 for Sessions 1, 2, 3). Key significant flows ($p<0.01$) included P3→CP1 and Fp1→CP1, plus a frontal-to-parietal link (Fz→Pz). LLM group (Sessions 1, 2, 3) showed progressively weaker connectivity, with Session 1 moderate, then declining by Session 3. These patterns imply that Session 4 (Brain-to-LLM group) engaged stronger attentional and memory processes.

Beta Band: Session 4 showed the highest connectivity sum (1.924) compared to Session 1 (1.656) and much lower in Sessions 2 and 3 (0.585, 0.275). Such beta band communication likely underlies active cognitive processing and sensorimotor integration; for instance, PO3→CP1 suggests visuomotor coordination (Figure 78). The elevated beta connectivity in Session 4 suggests that rewriting with AI possibly required higher executive and motor planning. LLM group's Session 1 also had substantial beta flows (perhaps due to initial tool adoption as well as task novelty), but these values dropped by Sessions 2 and 3.

Theta Band: Connectivity sums were 1.087 in Session 4 vs 0.394, 0.260, 0.132 in Sessions 1, 2, 3. Key theta flows (** $p<0.01$) included Pz→P4 and F3→Fp1, among others. These two links indicate engagement of frontoparietal working-memory networks. Theta activity is often linked to memory encoding and cognitive control; thus the pronounced theta connectivity in Session 4 suggests working memory load during rewriting. LLM group's theta connectivity was lower and diminished by Session 3 (Figure 77, right).

Delta Band: Delta connectivity was much larger in Session 4 (1.948) than in Sessions 1, 2, 3 (0.637, 0.408, 0.188). The strongest delta links ($p<0.01$) included O2→Fp1 and CP5→P4. For

example, the highly significant O2→Fp1 flow (p≈0.00013) indicates strong visual sensory influence on prefrontal regions, and CP5→P4 suggests cross-hemispheric integration. Delta-band interactions often reflect broad-scale cortical coupling. This may correspond to the intensive sensory-visual revision process when integrating AI-generated content. Notably, high frontal-temporal delta/theta coherence has been linked to poor writing performance in past studies [97], which may indicate the extra cognitive effort needed in Session 4. The LLM group's delta flows were weaker (Figure 77, left).

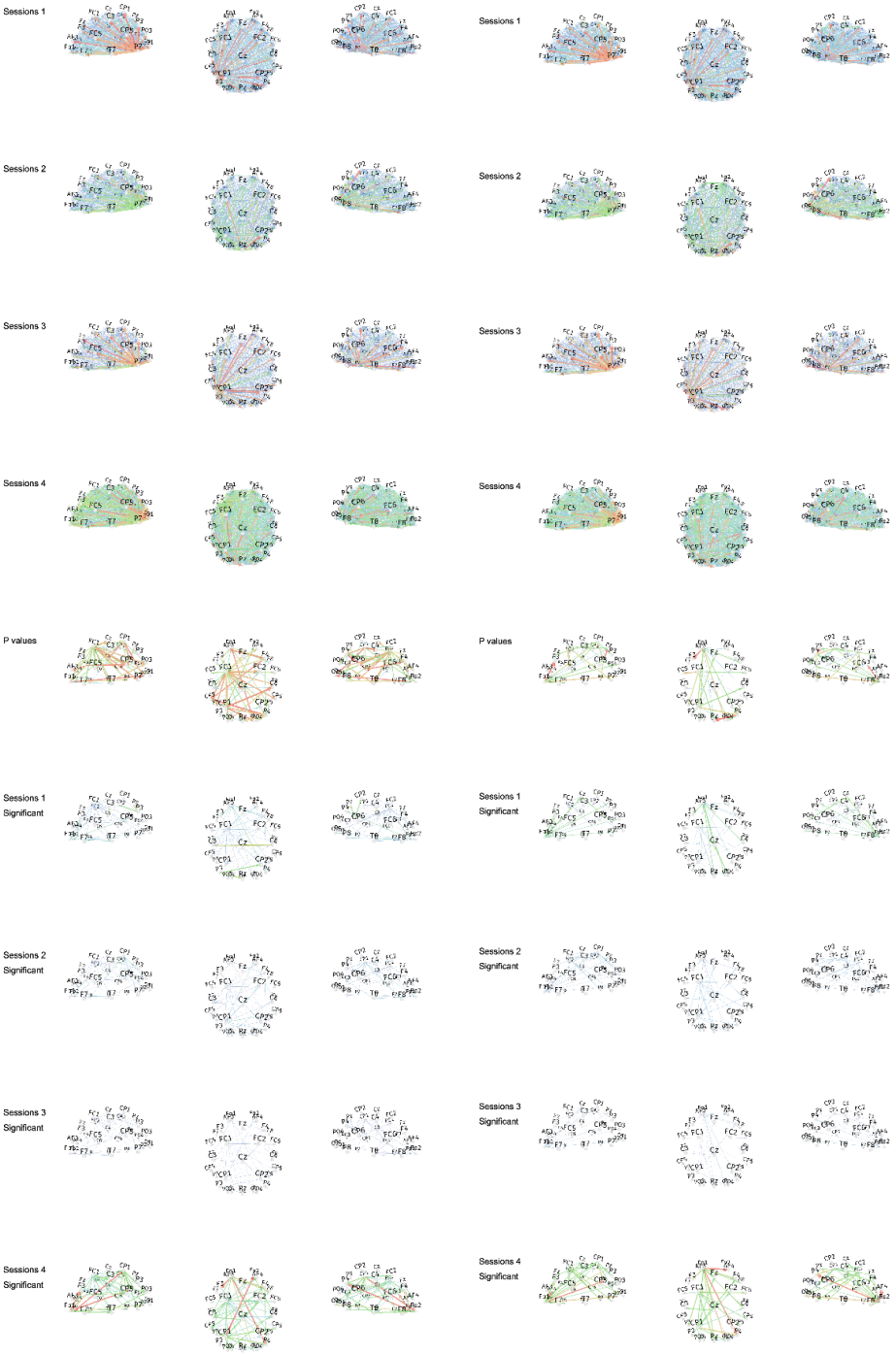




Figure 77. Dynamic Direct Transfer Function (dDTF) for Delta (left) and Theta (right) bands for LLM group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Interpretation

Across all frequency bands, Session 4 (Brain-to-LLM group) showed higher directed connectivity than LLM Group's sessions 1, 2, 3. This suggests that rewriting an essay using AI tools (after prior AI-free writing) engaged more extensive brain network interactions. One possible explanation is a novelty or cognitive load effect: Brain-to-LLM participants, encountering the LLM, needed to integrate its suggestions with existing knowledge, engaging multiple networks. In contrast, LLM Group had already adapted to using LLM tools by Session 1; their connectivity declined by Session 3, consistent with a neural efficiency adaptation, repeated practice leading to streamlined networks and less global synchrony. Such efficiency effects are known in skill learning: novices show widespread activation, experts, more focal processing [98, 99].

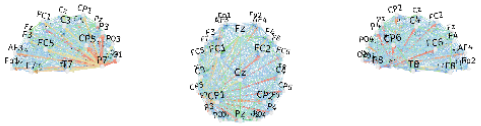
Band specific cognitive implications

The theta/alpha increases in Session 4 (especially in parietal and frontal regions) likely reflect greater involvement of attention and memory systems. Prior EEG studies found that parietal/central theta/alpha coherence supports memory encoding during writing, whereas excessive frontal delta/theta coherence signals difficulty [97]. Our Brain-to-LLM group's results (high theta/alpha flows) align with an increased memory retrieval and attentional demand. Beta connectivity increases suggest increases in sensorimotor and executive control processing, as discussed earlier. Beta band synchrony has been linked to active cognitive engagement and motor planning; the prevalent frontal→frontal and parietal→central beta flows possibly imply that participants were more actively monitoring and revising content. Delta connectivity may index deep cognitive integration of information across distant regions.

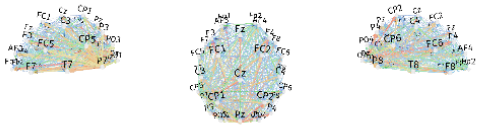
Inter-group differences: Cognitive Offloading and Decision-Making

The contrasting trends imply different neural mechanisms. LLM group's declining connectivity over sessions possibly suggests learning and network specialization with repeated AI tool use. Brain-to-LLM group's surge in connectivity at the first AI-assisted rewrite suggests that integrating AI output engages frontoparietal and visuomotor loops extensively. Functionally, AI tools may offload some cognitive processes but simultaneously introduce decision-making demands. The increased flows from parietal to central (e.g. P3→CP1) and occipital to frontal (O2→Fp1) in Session 4 most likely indicate that both spatial/visual processing and executive evaluation were upregulated.

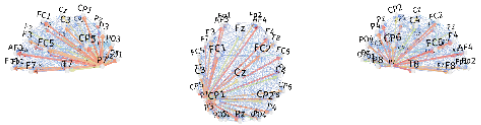
Sessions 1



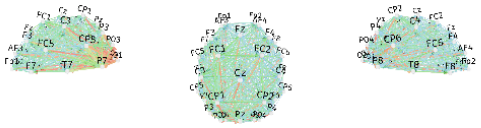
Sessions 2



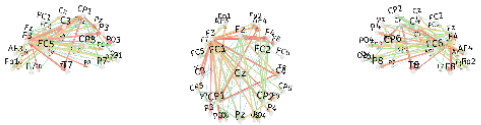
Sessions 3



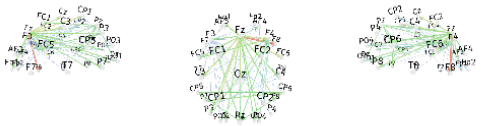
Sessions 4



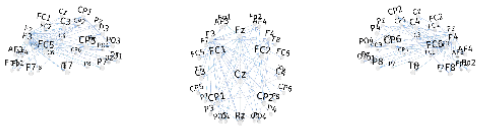
P values



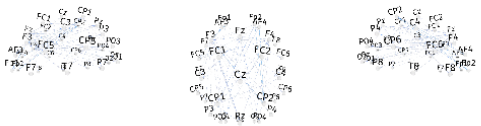
Sessions 1
Significant



Sessions 2
Significant



Sessions 3
Significant



Sessions 4
Significant

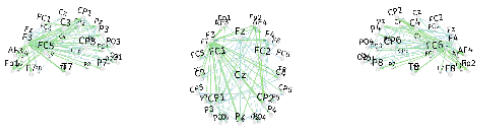




Figure 78. Dynamic Direct Transfer Function (dDTF) for Beta band for LLM group, and each of the sessions 1,2,3, 4. First four rows (session 1,2,3,4) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fifth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last four rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last four rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Neural Adaptation: from Endogenous to Hybrid Cognition in AI Assistance

Brain-to-LLM group entered Session 4 after three **AI-free** essays. The addition of AI assistance produced a network-wide spike in alpha-, beta-, theta-, and delta-band directed connectivity. Introducing exogenous suggestions into an endogenous workflow most likely forced the brain to reconcile internally stored plans with external prompts, increasing both attentional demand and integration overhead.

Task-switching studies show that shifting from one rule set to a novel one re-expands connectivity until a new routine is mastered [100]. Our data echoed this pattern: Brain-to-LLM group's first AI exposure re-engaged widespread occipito-parietal and prefrontal nodes, mirroring to an extent the frontoparietal 'initiate-and-adjust' control described in dual-network models of cognitive regulation [102].

In summary, AI-assisted rewriting after using no AI tools elicited significantly stronger directed EEG connectivity than initial writing-with-AI sessions. The group differences point to neural adaptation: LLM group appeared to have a reduced network usage, whereas novices from Brain-to-LLM group's recruited widespread connectivity when introduced to the tool.

TOPICS ANALYSIS

In-Depth NLP Topics Analysis Sessions 1, 2, 3 vs Session 4

Unlike previous n-grams analysis we did in the earlier section, here we expand into the n-grams with frequency 3 per essay, split between the sessions 1, 2, 3 and the 4th session.

The reader will find this analysis similar to the analysis present in Figure 27, which shows n-grams of order 4 and higher only **without the sessions separation**, therefore showing different aggregated results across the sessions, unlike this section, which shows n-grams use on the session level per topic.

We analyzed the most common n-grams per topic, group, session, with the n-grams that occurred at least 3 times in an essay. We observed several patterns, for example: for HAPPINESS topic (Figure 79) the Brain-only group used mostly 'true happiness' in session 1, however in session 4 (LLM-to-Brain) participants used 'I think' instead.

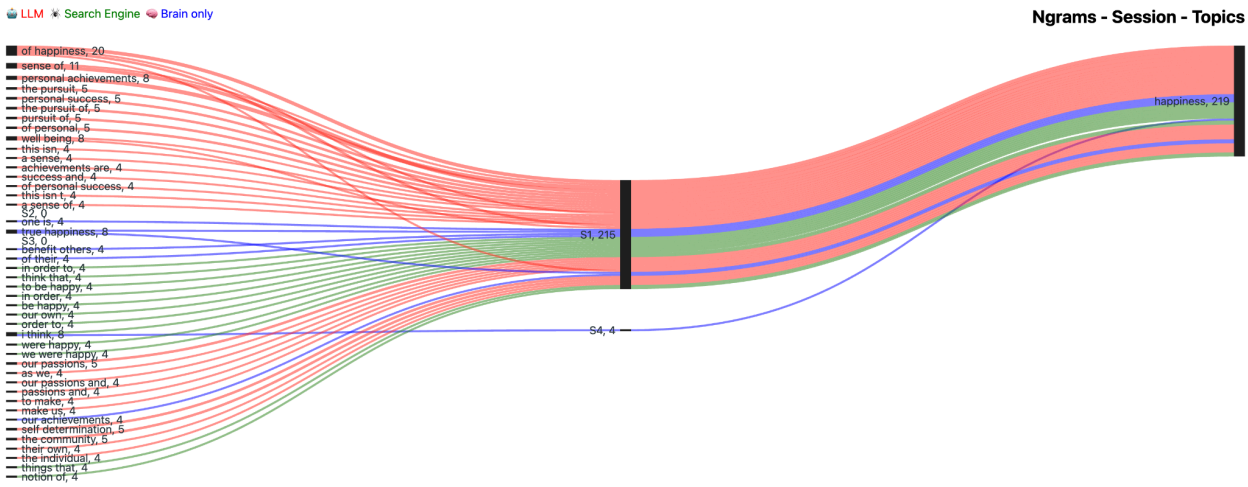


Figure 79. Frequency distribution of n-grams between different groups and sessions for topic HAPPINESS. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

In topic ART we can observe how LLM group used 'Matisse' n-gram quite frequently at first in session 2, however session 4 (Brain-to-LLM) did the opposite, and were more similar to a 'Brain-only' group, though they were using LLM (Figure 80). We can see that session 4 used the same n-grams as Brain and Search groups (majority of those were used in common): 'of art', 'works of', 'works of art'. In Appendix, Figure B we can see the neural connectivity differences in a participant while they were writing about topic ART.

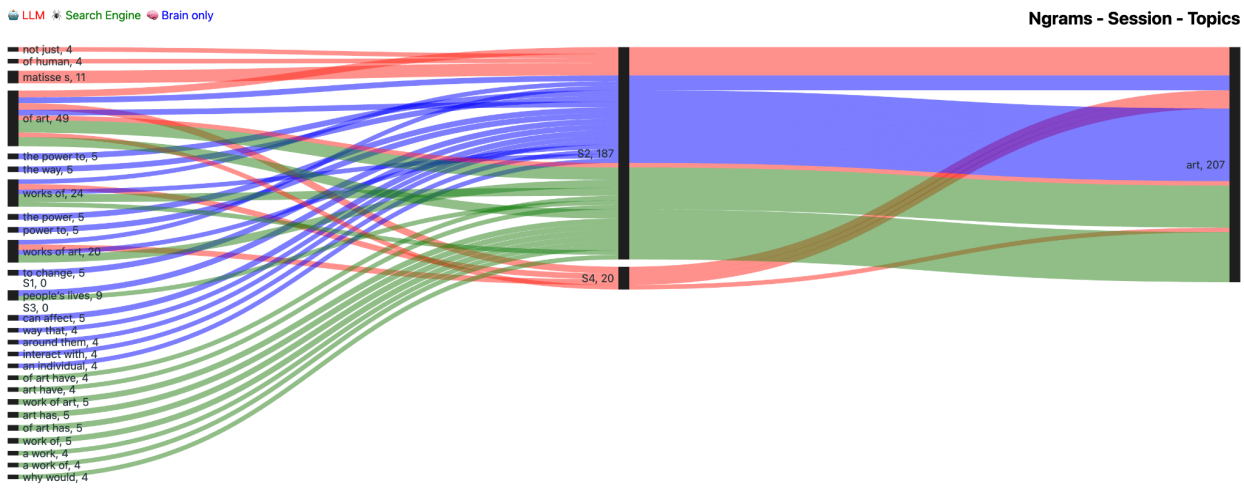


Figure 80. Frequency distribution of n-grams between different groups and sessions for topic ART. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

In the CHOICES topic (Figure 81) we can see a clear dominance of Brain-only group across both sessions 1 and 4. While LLM and Search Engine groups kept being repetitive (except the common n-grams like 'too many' or 'having too', etc.) the Brain-only group had a highly diverse range, where session 4 (LLM-to-Brain) focused on 'freedom'.

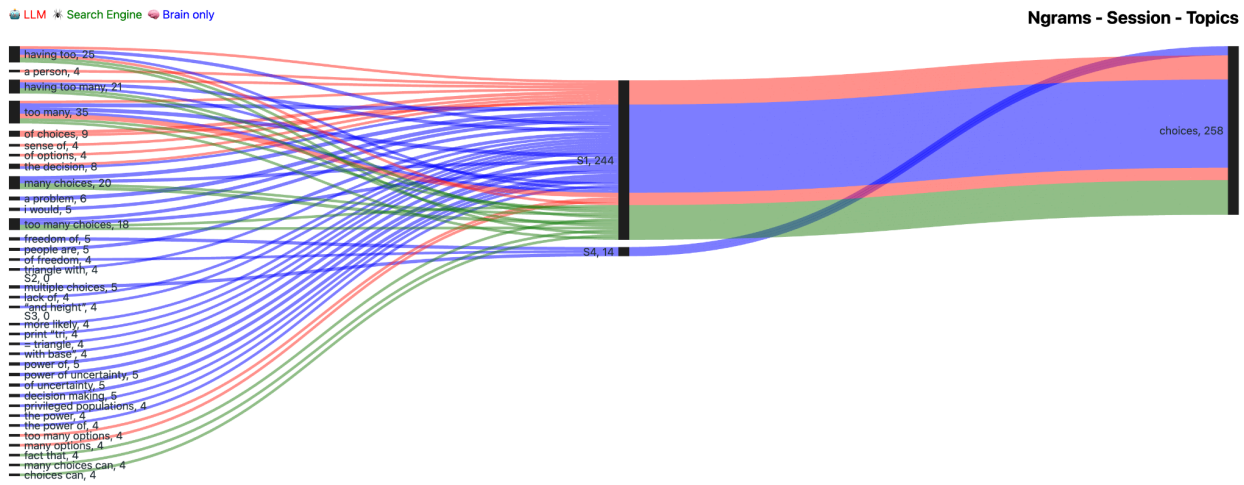


Figure 81. Frequency distribution of n-grams between different groups and sessions for topic CHOICES. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

For COURAGE topic, most participants used 'to show' n-gram, however Session 4 had a different behaviour: where LLM-to-Brain group (in blue) reused same 'to show' n-gram, likely remembering the previously written topic, as well as 'hard to' n-gram, however Brain-to-LLM group used 'being vulnerable' n-gram (Figure 82).

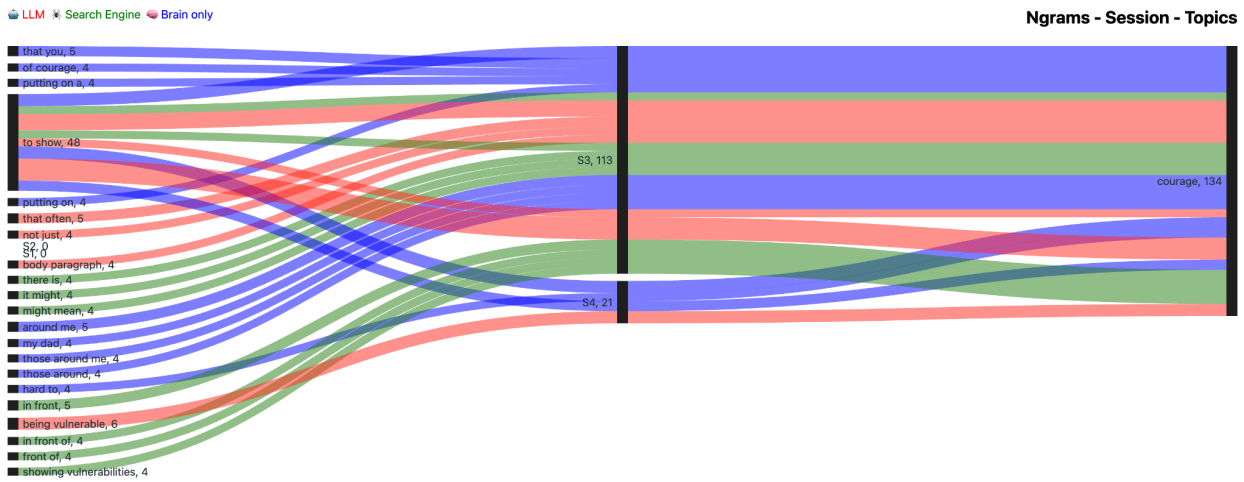


Figure 82. Frequency distribution of n-grams between different groups and sessions for topic COURAGE. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

In the FORETHOUGHT topic the Brain-only group participants occasionally used 'think twice' n-gram (Figure 83). And Session 4, LLM-to-Brain group (blue) again showed how participants reused 'before speaking' n-gram, which was actively used by LLM group before in session 2 (red arrow). Also in Appendix, Figure B we can see the differences in the participant's neural connectivity while they were writing about the topic FORETHOUGHT.

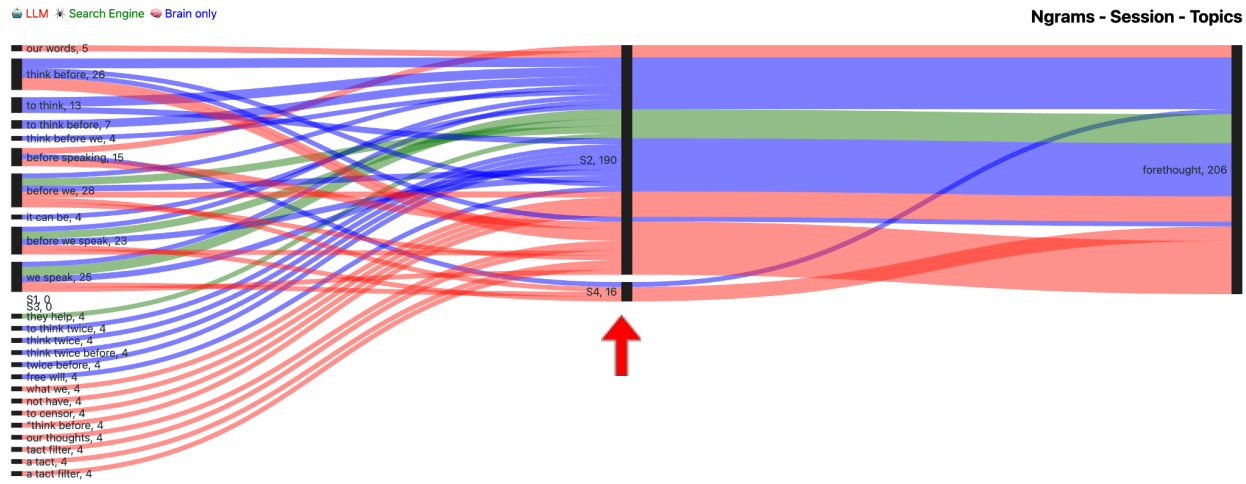


Figure 83. Frequency distribution of n-grams between different groups and sessions for topic FORETHOUGHT. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue). Red arrow points up to LLM-to-Brain (blue) reuse of 'think before' that is actively used by LLM before.

In the LOYALTY topic (Figure 84) the Brain-only group stood out by using 'true loyalty' n-gram, where LLM somehow managed to talk about 'colorado springs'.

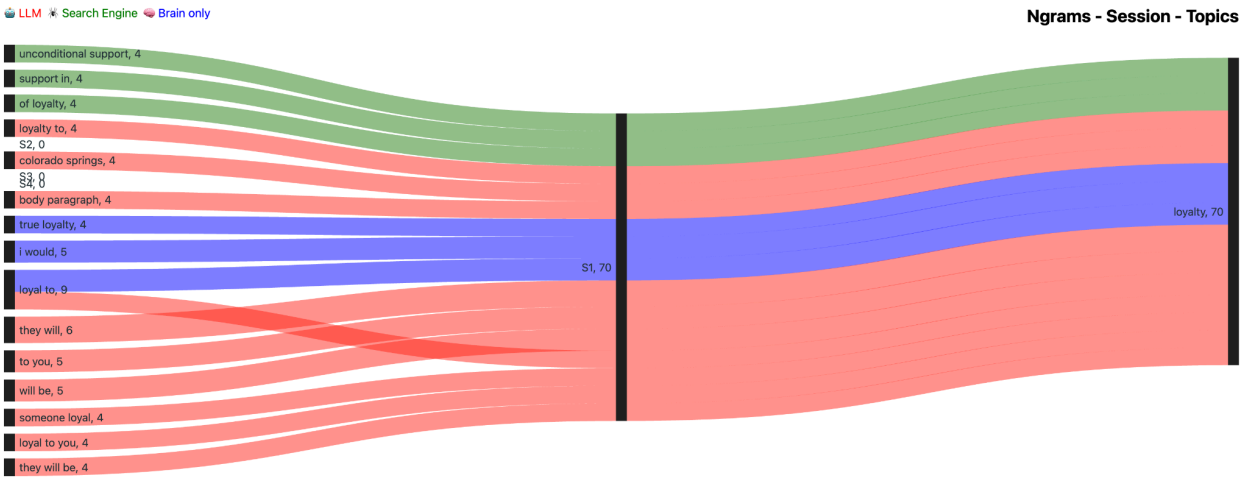


Figure 84. Frequency distribution of n-grams between different groups and sessions for topic LOYALTY. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

The PERFECT topic (Figure 85) carried similar pattern for session 4 LLM-to-Brain by reusing some of the n-grams like "perfect' society' with the quotes, demonstrating same pattern as LLM group in session 3, however Brain-to-LLM group in session 4 validated dominance of the LLM n-grams like 'perfect society' hinting that participants may have leaned on the model's suggested phrasing with relatively little further revision.

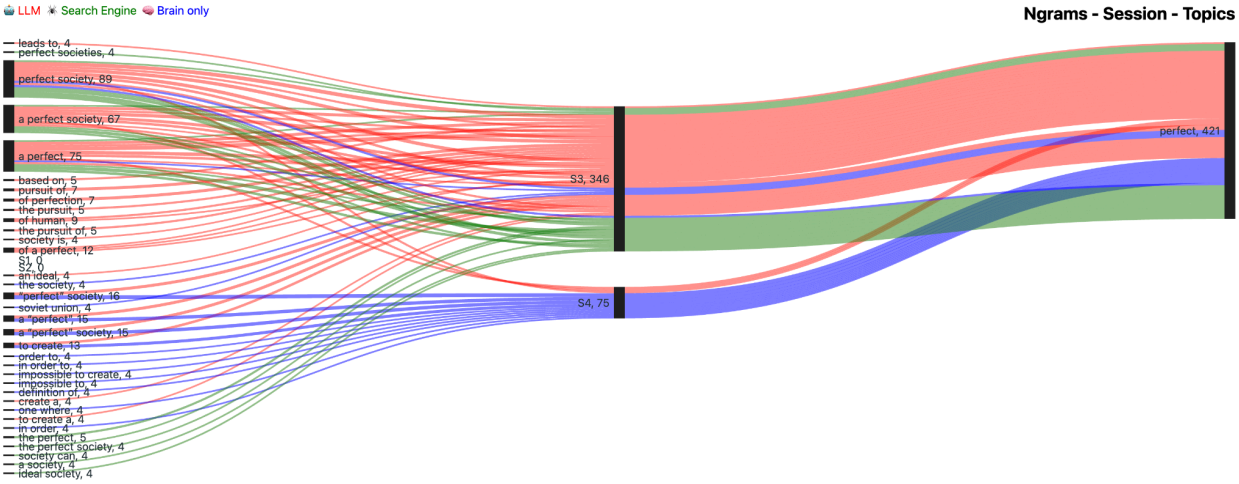


Figure 85. Frequency distribution of n-grams between different groups and sessions for topic PERFECT. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

In the PHILANTHROPY topic we can see the impact of 'homeless person' n-gram on the frequent use in the Search group (Figure 86).

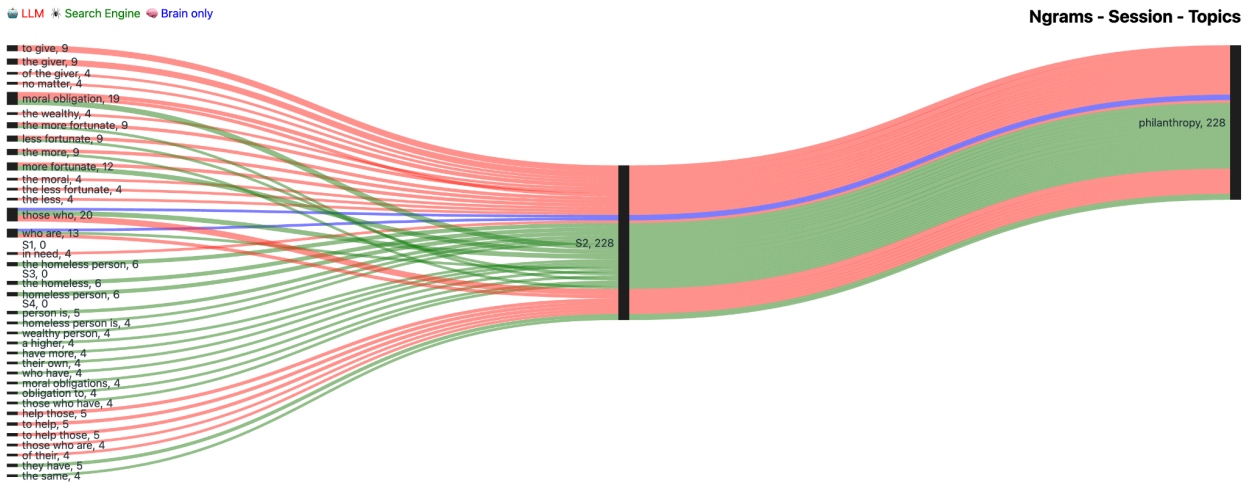


Figure 86. Frequency distribution of n-grams between different groups and sessions for topic PHILANTHROPY. Left column includes n-grams. Middle column shows sessions, and the last column specifies the topic. Color lines demonstrate what tools were used: LLM (red), Search Engine (green), Brain-only (blue).

To summarize the findings, different groups clearly had different frequency patterns for n-grams across the topics. Session 4 had two distinct groups: Brain-to-LLM and LLM-to-Brain. Brain-to-LLM group in session 4 gave in to LLM suggestions in the essay writing, and LLM-to-Brain group seemed to have suffered from the previous LLM bias, and kept reusing same vocabulary and structure, when Brain-only group in Sessions 1,2,3 did not. However, as the number of participants recorded in session 4 was 18, this analysis requires further data collection from a wider population to draw the definite conclusions.

Neural and Linguistic Correlates on the Topic of Happiness

LLM Group

Participants in the LLM group, who used LLMs during the essay-writing task, exhibited a distinct linguistic and neural profile. The most frequent n-grams in their essays were '*choos career*' (see Figure 25, top-right red circle with frequency of 4) and '*person success*,' (Figure 27), indicating a focus on individual ambition and achievement.

The LLM group demonstrated the lowest overall dDTF across all frequency bands, with especially diminished activity in the Alpha (Figure 89) and Theta bands networks (Figure 87) commonly associated with attentional control, semantic integration, and internal reflection as mentioned in the previous section on EEG analysis. Connectivity patterns revealed weak engagement in frontoparietal and prefrontal pathways, notably between FC1 and Fp1, and F7 and F3. These reduced functional connections suggest limited recruitment of regions involved in higher-order cognitive functions such as goal maintenance, moral reasoning, and emotionally grounded decision-making.

These n-grams suggest goal-oriented phrasing that aligns with generic success narratives often found in LLM-generated text. The minimal connectivity, particularly in frontal and semantic hubs

(e.g. AF3, F3), supports the hypothesis that the tool generated much of the language, and the user exerted little integration or reflection.

Altogether, the neural and linguistic evidence points toward a more externally scaffolded writing process with minimal reliance on endogenous semantic or affective regulation, potentially reflecting the influence of tool-driven composition over self-generated reflection.

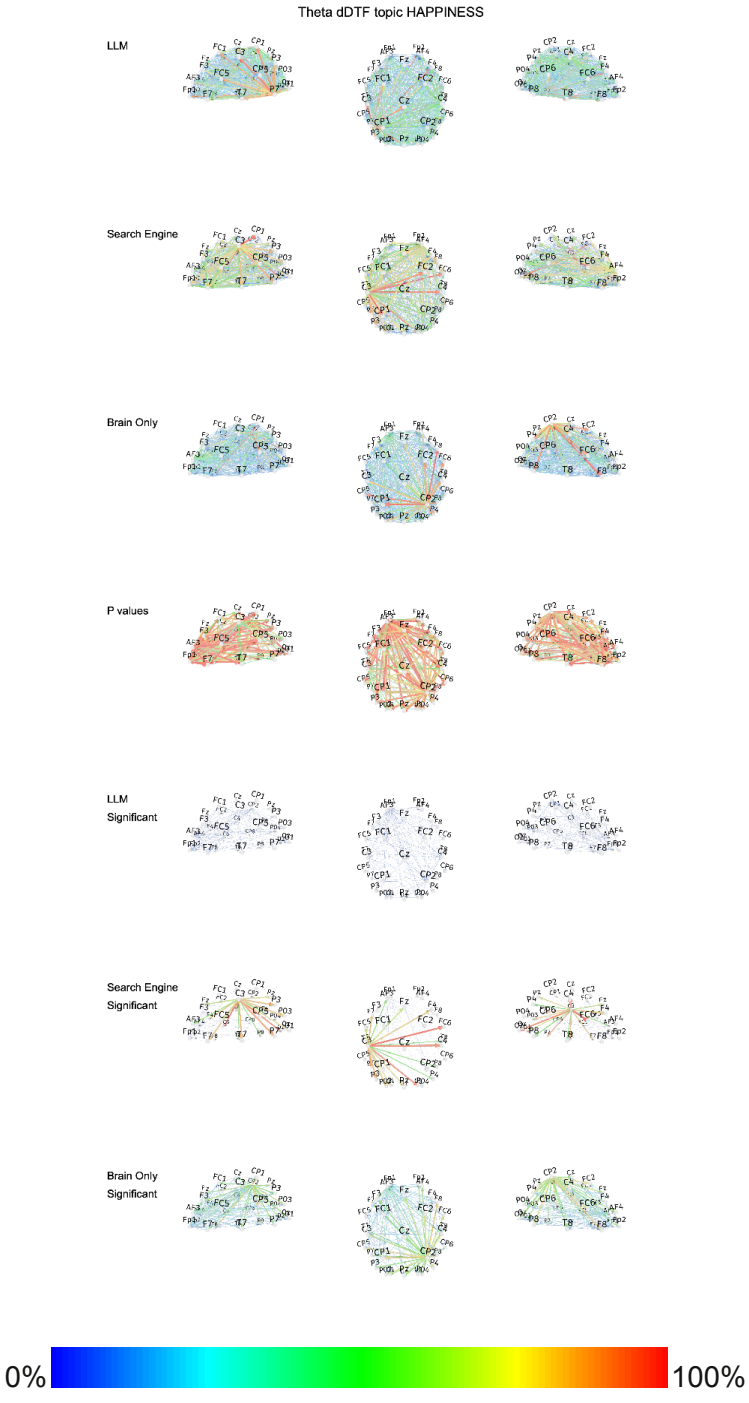


Figure 87. Dynamic Direct Transfer Function (dDTF) for Theta band for Happiness topic between all groups. Rows 1 (LLM group), 2 (Search Engine group), and 3 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fourth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last three rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last three rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Search Group

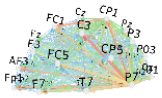
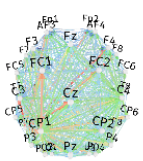
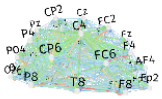
Participants in the Search Engine group, who used a search engine during the essay-writing task, also showed a distinctive linguistic and neural profile. The top n-gram in their essays '*give us*' (see Figure 25, green circle with frequency of 4) suggests a more outward-facing rhetorical style, possibly reflecting appeals to collective values or external authority.

This group exhibited elevated delta and high-delta dDTF connectivity (Figure 88), with notable inflows targeting C3, Fp1, and AF4. These patterns are indicative of increased bottom-up processing, suggesting that participants were actively integrating externally retrieved information under conditions of heightened cognitive effort. The connectivity profile implies a reliance on externally sourced material, processed through more effortful semantic and attentional pathways.

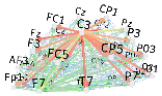
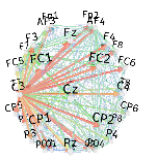
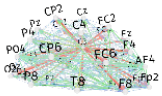
The phrase 'give us' implied passive framing, possibly reflecting external sourcing (e.g. quoting or summarizing from online texts). This likely aligned with their delta band increase, often linked to external attention, monitoring, or effortful stimulus integration.

Delta dDTF topic HAPPINESS

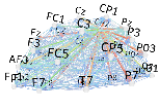
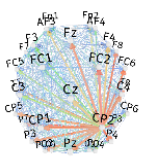
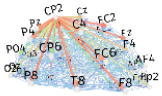
LLM

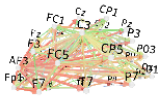
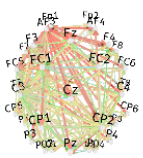
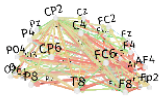
Search Engine

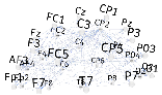
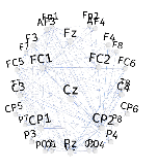
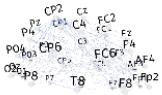
Brain Only

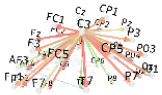
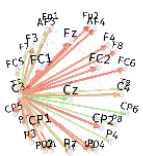
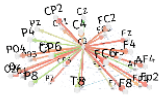
P values

LLM
Significant

Search Engine
Significant

Brain Only
Significant

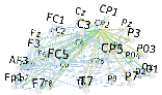
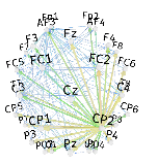
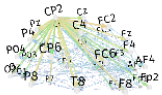






Figure 88. Dynamic Direct Transfer Function (dDTF) for Delta band for Happiness topic between all groups. Rows 1 (LLM group), 2 (Search Engine group), and 3 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fourth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last three rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last three rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Brain-only Group

Participants in the Brain-only group, who completed the essay task without any external tools, used n-grams such as '*true happi*' and '*benefit other*'. Their EEG data showed the highest dDTF connectivity across all frequency bands, with particularly robust directional coupling from frontal to parietal regions and from visual to prefrontal areas (e.g. FC1→Fp1, Oz→AF3). This pattern suggests engagement in internally driven, emotionally grounded reasoning, likely involving abstract thought and self-regulation in the absence of external cognitive scaffolding.

These phrases reflect reflective and prosocial framing markers of internally-driven semantic processing. The elevated connectivity in frontal, parietal, and limbic-associated areas supports the notion of deep cognitive-emotional integration, likely necessary for values-based arguments.

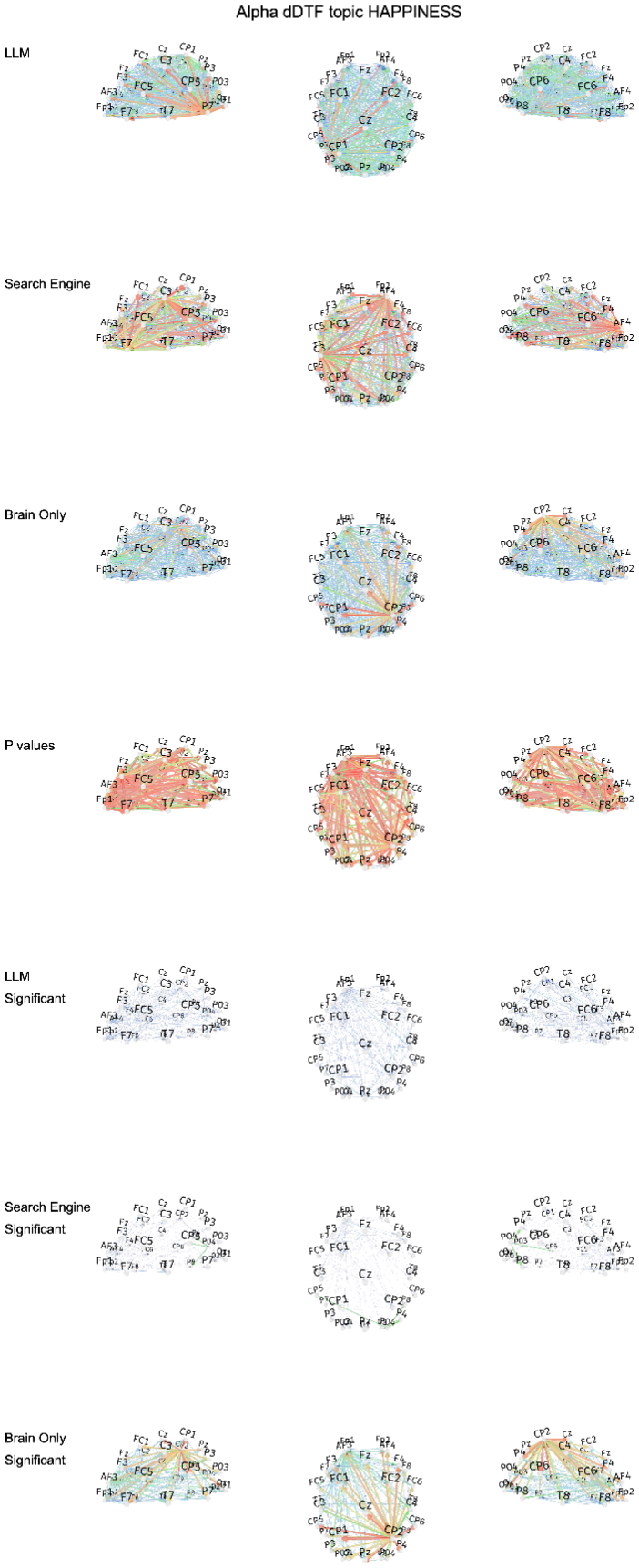

Alpha dDTF topic HAPPINESS
LLM
Search Engine
Brain Only
P values
LLM
Significant
Search Engine
Significant
Brain Only
Significant



Figure 89. Dynamic Direct Transfer Function (dDTF) for Alpha band for Happiness topic between all groups. Rows 1 (LLM group), 2 (Search Engine group), and 3 (Brain-only group) show the dDTF for all pairs of 32 electrodes = 992 total. Blue is the lowest dDTF value, red is the highest dDTF value. Fourth row (P values) shows only significant pairs, where red ones are the most significant and blue ones are the least significant (but still below 0.05 threshold). Last three rows show only significant dDTF values filtered using the third row of p values, and normalized by the min and max ones in the last three rows. Thinnest blue lines represent significant but weak dDTF values, and red thick lines represent significant and strong dDTF values.

Though this analysis remains speculative, as we only performed it for one topic, a relationship seems to emerge between the provenance of n-grams and the brain's connectivity patterns across groups. Participants who generated more abstract, introspective, or value-oriented phrases exhibited stronger intrinsic neural coupling, whereas those who depended on external aids, whether LLMs or search engines, tended to produce more generic, outwardly framed statements that align with reduced cognitive integration. In summary, these observations might indicate that the choice of tool (or its absence) not only shaped neural dynamics but also steered participants toward particular concepts and linguistic forms.

DISCUSSION

The results of our study offer several intriguing insights into the differences in cognitive and performance outcomes in essay writing tasks for 54 participants, who used LLMs such as ChatGPT, traditional web search, or were tools-free over a span of 4 sessions per participant over a period of 4 months.

NLP

We found that the Brain-only group exhibited strong variability in how participants approached essay writing across most topics. In contrast, the LLM group produced statistically homogeneous essays within each topic, showing significantly less deviation compared to the other groups. The Search Engine group was likely, at least in part, influenced by the content that was promoted and optimized by a search engine (see Figure 90 below for PHILANTHROPY topic keywords), therefore, the keywords used to promote specific ideas within each topic were likely influenced more by the participants' own queries than by the prompts provided in the LLM group. Interestingly, in the Brain-only group the social media influence found its way around, here is a quote from one of the essays '*So why we are not talking about it on Instagram, for example?*'.

Keyword (by relevance)	Avg. monthly searches	Three month change	YoY change	Competition	Ad impression share	Top of page bid (low range)	Top of page bid (high range)
Keywords you provided							
giving	10K – 100K	0%	0%	Low	–	$1.76	$7.00
homeless	10K – 100K	+900%	+900%	Low	–	$0.56	$6.87
philanthropy	100K – 1M	0%	0%	Low	–	$1.34	$6.59
charities	10K – 100K	+900%	+900%	Low	–	$5.00	$22.94
Keyword ideas							
st jude donation	10K – 100K	0%	0%	High	–	$5.89	$19.14

Figure 90. Google Ads Keywords planner shows AI suggested bidding based on the real-time demand and supply. Higher price means higher demand. 'Keywords you provided' section demonstrates preselected keywords for the price and audience breakdown. June 8, 2025.

In our NLP analysis we discovered that the LLM group used the most of the specific named entities (NERs) such as persons, names, places, years, definitions, while the Search Engine group used at least two times less NERs, and the Brain-only group used 60% less of NERs compared to the LLM group.

Across the essays written by different groups one can likely observe propagation of biases used in the training data of the used LLM (Figure 91), or advertisements in a search engine (see Figure 90), or human biases, like getting an education in the same environment but with

different (Figure 2, Figure 3) cultural, linguistic, and other backgrounds. The prompts and queries used by participants (Figure 33), sequentially impacted how participants structured ontology and semantics of the essays. Few participants relied less on the LLM's 'opinion' (bias) in the topics like PHILANTHROPY and FORETHOUGHT, and in other topics, like ART and PERFECT, participants behaved differently, based on the analysed prompts and interviews. Interestingly, several participants used languages other than English (Spanish, Portuguese), but eventually ended up with the English essays that were not very different from others within the same LLM group and same topic. The Search Engine group participants were more prone to experience the filter bubble [108] in their search results (Figure 92).

Participants in the LLM and Search Engine groups were more inclined to focus on the output of the tools they were using because of the added pressure of limited time (20 minutes). Most of them focused on reusing the tools' output, therefore staying focused on copying and pasting content, rather than incorporating their own original thoughts and editing those with their own perspectives and their own experiences.

In the lexical n-gram analysis (Figure 25) we found that LLM had a bias of higher probability of third-person address forms [114] (see Figure 91) and focusing on career aspects ('choos career') (see Figure 27).

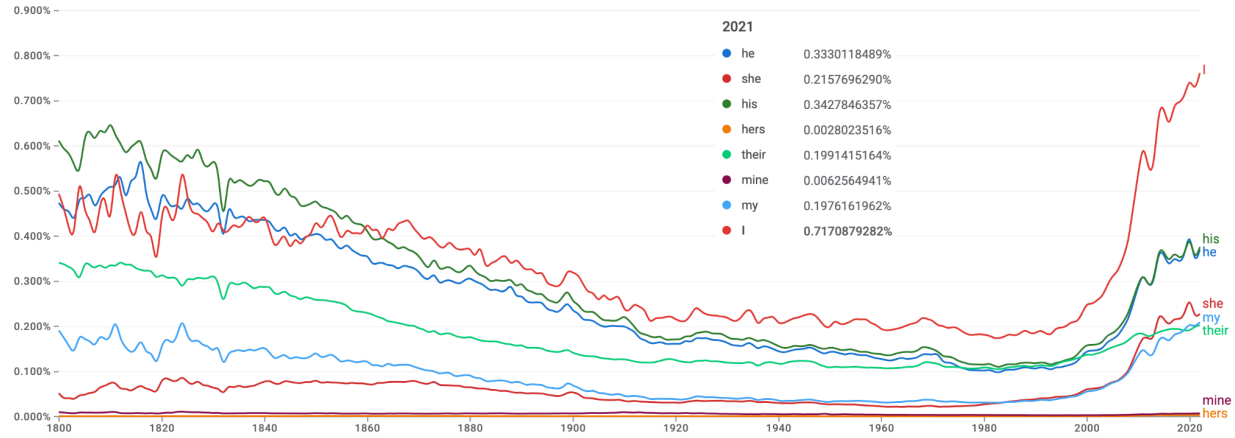


Figure 91. Probability of n-grams in the published literature according to Google N-gram viewer in books published from 1800 to 2021 (the books subset used to train OpenAI ChatGPT).

If we look at the n-grams and topics more closely, for example topic PHILANTHROPY (Figure 86), we can see that the Search Engine group was heavily leaning into using 'homeless' based n-grams, however the LLM group is focused around the 'giving' aspect in the n-grams. According to Google Keywords Planner data (Figure 90), we can see the bid size around $7 per ad placement for both 'giving' and 'homeless'. However 'homeless' has almost a 900% increase in monthly searches compared to 'giving'. Same for 'charities', but the bid price for 'charities' is triple, around $23 per ad placement. And we can see in Figure 92 the trending of 'giving' is much higher across the Google Search according to Google Trends. It is likely that the Search Engine group experienced a bias from the tool, and was susceptible to the tool's output.

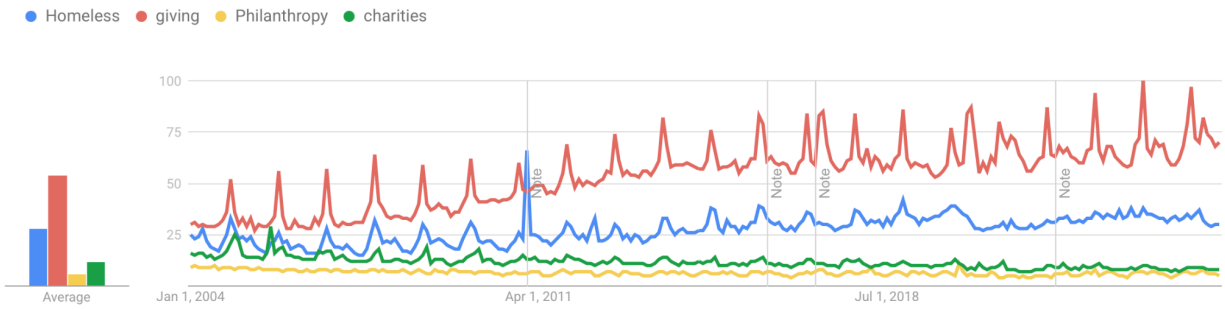


Figure 92. Homeless vs Giving vs Philanthropy vs Charities in Google Trends data from 2004 to 2024.

The ontology (see Agent's prompt structure in Figure 34) analysis demonstrated significant correlation between the LLM group and the Search Engine group, with almost no intersection with the essays written by the Brain-only group within the same topics as well as between all topics. Interestingly, the Brain-only group touched more on the freedom/liberty parts, while the Search Engine and the LLM groups focused more on the justice aspects (Figure 37). It is worth noting that the LLM group [126,127] focused heavily on linking the Art topic to its objective aspects (what art is being applied to), whereas the Search Engine group emphasized its subjective dimensions (who is creating the art).

We created an AI judge to leverage scoring and assessments in the multi-shot fine-tuning (Figure 39) based on the chosen topics, and we also asked human teachers to do the same type of scoring the AI judge did. Human teachers were already exposed in their day-to-day work to the essays that were written with the help of LLMs, therefore they were much more sceptical about uniqueness and content structure, whereas the AI judge consistently scored essays higher in the uniqueness and quality metrics. The human teachers pointed out how many essays used similar structure and approach (as a reminder, they were not provided with any details pertaining to the conditions or group assignments of the participants). In the top-scoring essays, human teachers were able to recognize a distinctive writing style associated with the LLM group (independent of the topic), as well as topic-specific styles developed by participants in both the LLM and Search Engine groups (see Figure 37). Interestingly, human teachers identified certain stylistic elements that were consistent across essays written by the same participant, often attributable to their work experience. In contrast, the AI judge failed to make such attributions, even after multi-shot fine-tuning and projecting all essays into a shared latent vector space.

Neural Connectivity Patterns

EEG analysis presented robust evidence that distinct modes of essay composition produced clearly different neural connectivity patterns, reflecting divergent cognitive strategies (Figure 1). Dynamic Directed Transfer Function (dDTF) analysis revealed systematic and frequency-specific variations in network coherence, with implications for executive function, semantic processing, and attention regulation.

Brain connectivity systematically scaled down with the amount of external support: the Brain-only group exhibited the strongest, widest-ranging networks, Search Engine group showed intermediate engagement, and LLM assistance elicited the weakest overall coupling. Activations and connectivity were the most prominent in the Brain-Only group, which consistently exhibited the **highest total dDTF connectivity across alpha, theta, and delta bands**, particularly in temporo-parietal and frontal executive regions. This was followed by the Search Engine group, which demonstrated **approximately 34-48% lower total connectivity** across the brain depending on frequency band, especially in lower frequencies. The **LLM group** showed the **least extensive connectivity**, with up to **55% reduced total dDTF magnitude** compared to the Brain-Only group in low-frequency semantic and monitoring networks.

Interestingly, the Search Engine group exhibited **increased activity in the occipital and visual cortices**, particularly in alpha and high alpha sub-bands. This pattern most likely reflects the group's engagement with visually acquired information during the research and content-gathering phase during the use of the web browser. These occipital-to-frontal flows (e.g. Oz→Fp2, PO4→AF3) support the interpretation that participants were actively scanning, selecting, and evaluating information presented on the screen to construct their essays, a cognitively demanding integration of visual, attentional, and executive resources.

In contrast, despite also using a digital interface, the LLM group did **not exhibit comparable levels of visual cortical activation**. While participants interacted with the LLM via a screen, the purpose of this interaction was distinct: LLM use reduced the need for prolonged visual search and semantic filtering, shifting cognitive load toward procedural integration and motor coordination (e.g. FC6→CP5, Fp1→Pz), as supported by dominant beta band activity in fronto-parietal networks. This suggests a more **automated, scaffolded cognitive mode**, with reduced reliance on endogenous semantic construction or visual content evaluation.

Meanwhile, the Brain-only group showed the **strongest activations outside of the visual cortex**, particularly in **left parietal, right temporal, and anterior frontal areas** (e.g. P7→T8, T7→AF3). These regions are involved in semantic integration, creative ideation, and executive self-monitoring. The elevated delta and theta coherence into AF3, a known site for cognitive control, underscored the high internal demand for content generation, planning, and revision in the absence of external aids.

Collectively, these findings support the view that external support tools restructure not only task performance but also the underlying cognitive architecture. The Brain-only group leveraged broad, distributed neural networks for internally generated content; the Search Engine group relied on hybrid strategies of visual information management and regulatory control; and the LLM group optimized for procedural integration of AI-generated suggestions.

These distinctions carry significant implications for cognitive load theory, the extended mind hypothesis [102], and educational practice. As reliance on AI tools increases, careful attention must be paid to how such systems affect neurocognitive development, especially the potential trade-offs between external support and internal synthesis.

Behavioral Correlates of Neural Connectivity Patterns

The behavioral data, particularly around quoting ability, correctness of quotes, and essay ownership, supports our neural connectivity findings. These results suggest that the functional network dynamics engaged during essay writing not only predicted but also shaped cognitive processes, including the consolidation of memory traces, efficiency of self-monitoring, and the degree of perceived agency over the written work.

Quoting Ability and Memory Encoding

The most consistent and significant behavioral divergence between the groups was observed in the ability to quote one's own essay. LLM users significantly underperformed in this domain, with 83% of participants (15/18) reporting difficulty quoting in Session 1, and none providing correct quotes. This impairment persisted albeit attenuated in subsequent sessions, with 6 out of 18 participants still failing to quote correctly by Session 3.

This difficulty maps directly onto the reduced low-frequency connectivity in LLM group, particularly in the theta (4-8 Hz) and alpha (8-12 Hz) bands, which are heavily implicated in episodic memory consolidation and semantic encoding [84, 103, 104]. These oscillations are typically strongest when individuals generate and internally structure content, rather than passively integrating externally generated information. The reduced dDTF strength in frontal and temporal nodes among LLM users likely reflected a bypass of deep memory encoding processes, whereby participants read, selected, and transcribed tool-generated suggestions without integrating them into episodic memory networks.

Search Engine and Brain-only participants did not display such impairments. By Session 2, both groups achieved near-perfect quoting ability, and by Session 3, 100% of both groups' participants reported the ability to quote their essays, with only minor deviations in quoting accuracy. This behavioral preservation correlates with stronger parietal-frontal and temporal-frontal connectivity in alpha and theta bands, observed especially in the Brain-only group, and to a lesser degree in the Search Engine group. In the Brain-only group, the P7→T8 and Pz→T8 connections suggest deep semantic processing, while Oz→Fz and FC6→AF3 reflect sustained executive monitoring, both of which support stronger integration of content into memory systems.

Correct Quoting

Correct quoting ability, which goes beyond simple recall to reflect semantic precision, showed the same hierarchical pattern: Brain-only group > Search Engine group > LLM group. The complete absence of correct quoting in the LLM group during Session 1, and persistent impairments in later sessions, suggested that not only was memory encoding shallow, but the semantic content itself may not have been fully internalized.

This lack of quote correctness underscores the reduced frontal-temporal semantic coherence in LLM group, particularly the near-absence of T7/8-targeted pathways, a region crucial for verbal and conceptual integration [105]. In contrast, there was a strong convergence on T8 and AF3 in the Brain-only group.

Essay Ownership and Cognitive Agency

Another nuanced behavioral dimension was the participants' perception of essay ownership. While Brain-only group claimed full ownership of their texts almost unanimously (16/18 in Session 1, rising to 17/18 by Session 3), LLM Group presented a fragmented and conflicted sense of authorship: some participants claimed full ownership, others explicitly denied it, and many assigned partial credit to themselves (e.g. between 50-90%).

These responses suggest a diminished sense of cognitive agency. From a neural standpoint, this aligns with the reduced convergence on anterior frontal regions (AF3, Fp2), which are involved in error monitoring, and self-evaluation [106]. In the LLM group, the delegation of content generation to external systems appeared to have disrupted these metacognitive loops, resulting in a psychological dissociation from the written output.

The Search Engine group, which relied on the web browser, showed more stable ownership patterns but still less certainty than the Brain-only group. Participants often reported partial authorship (e.g. 70-90%), likely due to the interleaving of internal synthesis with external retrieval, a cognitive process supported by their posterior-frontal alpha and delta connectivity.

Cognitive Load, Learning Outcomes, and Design Implications

Taken together, the behavioral data revealed that higher levels of neural connectivity and internal content generation in the Brain-only group correlated with stronger memory, greater semantic accuracy, and firmer ownership of written work. Brain-only group, though under greater cognitive load, demonstrated deeper learning outcomes and stronger identity with their output. The Search Engine group displayed moderate internalization, likely balancing effort with outcome. The LLM group, while benefiting from tool efficiency, showed weaker memory traces, reduced self-monitoring, and fragmented authorship.

This trade-off highlights an important educational concern: AI tools, while valuable for supporting performance, may unintentionally hinder deep cognitive processing, retention, and authentic engagement with written material. If users rely heavily on AI tools, they may achieve superficial fluency but fail to internalize the knowledge or feel a sense of ownership over it.

Session 4

Our dDTF analysis revealed that Session 4, which included the participants who came from the original LLM group, the so-called LLM-to-Brain group, produced a distinctive neural connectivity profile that was significantly different from progression patterns observed in Sessions 1, 2, 3 in

the Brain-only group. While these LLM-to-Brain participants demonstrated substantial improvements over 'initial' performance (Session 1) of Brain-only group, achieving significantly higher connectivity across frequency bands, they consistently underperformed relative to Session 2 of Brain-only group, and failed to develop the consolidation networks present in Session 3 of Brain-only group. Original LLM participants might have gained in the initial skill acquisition using LLM for a task, but it did not substitute for the deeper neural integration, which can be observed for the original Brain-only group. Educational interventions should consider combining AI tool assistance with tools-free learning phases to optimize both immediate skill transfer and long-term neural development. The absence of highly significant connections ($p < 0.001$) in Session 4 for original LLM group's participants, indicates potential limitations in achieving robust neural synchronization essential for complex cognitive tasks. The preserved FC5-centered networks indicated that AI tools established basic motor coordination, but the missing frontal-to-parietal executive networks suggest the need for additional cognitive training components.

Regarding Session 4 participants, those who had previously written without tools (Brain-only group), the so-called Brain-to-LLM group, exhibited significant increase in brain connectivity across all EEG frequency bands when allowed to use an LLM on a familiar topic. This suggests that AI-supported re-engagement invoked high levels of cognitive integration, memory reactivation, and top-down control. By contrast, repeated LLM usage across Sessions 1, 2, 3 for the original LLM group reflected reduced connectivity over time. These results emphasize the dynamic interplay between cognitive scaffolding and neural engagement in AI-supported learning contexts.

Regarding Session 4, which included the participants who came from the original Brain-only group, from an educational standpoint, these results suggest that strategic timing of AI tool introduction following initial self-driven effort may enhance engagement and neural integration. The corresponding EEG markers indicate this may be a more neurocognitively optimal sequence than consistent AI tool usage from the outset.

We interviewed all participants after the essay writing and asked them to reflect on the tools usage, and asked them to explain what they wrote about and why. With most participants in the Brain-only group engaging and caring more about 'what' they wrote, and also 'why' (see Figure 32, where participants in Session 4 used 'information seeking' prompts 3 times more often than in sessions 1, 2, 3), while the other groups briefly focused on the 'how' part. During the 4th session, when we asked participants to pick the topic, but use an opposite tool, the participants who used no tools before, performed more fine-tuned prompts when they used LLM tools, similar to how the Search Engine group used to compose queries in their search. Though those participants who used LLM tools in the previous session, mostly wrote a different or a deeper version of the essays in the 4th session.

Behavioral Correlates of Neural Connectivity Patterns in Session 4

In Session 4, removing AI support significantly impaired the participants from original LLM group: 78 % failed to quote anything (Question 5) and only 11 % were able to produce a correct quote (Question 6), compared with 11 % and 78 % in the Brain-only Group.

Neurophysiological data in part explained this impairment. dDTF analysis revealed that LLM-to-Brain group lacked the robust fronto-parietal synchronization (e.g. Fz→P4, AF3→CP6) normally associated with deep semantic encoding and source-memory retrieval, processes essential for accurate quotation [107]. Moreover, the LLM-to-Brain participants showed no high-significance connectivity clusters ($p < 0.001$), pointing to attenuated neural connectivity during retrieval. Although isolated FC5-centered motor networks were still present, consistent with preserved typing routine, such activity was insufficient to compensate for reduced semantic recall. In contrast, Brain-to-LLM participants (from original Brain-only group) displayed stronger dDTF magnitudes across frontal, temporal, and occipital pathways, reflecting effective top-down regulation, episodic access, and re-encoding that aligned with their superior behavioral accuracy. These converging findings thus suggest that habitual LLM support might potentially compromise the behavioral competence required for quoting.

This correlation between neural connectivity and behavioral quoting failure in LLM group's participants offers evidence that:

1. **Early AI reliance may result in shallow encoding.**
   LLM group's poor recall and incorrect quoting is a possible indicator that their earlier essays were not internally integrated, likely due to outsourced cognitive processing to the LLM.
2. **Withholding LLM tools during early stages might support memory formation.**
   Brain-only group's stronger behavioral recall, supported by more robust EEG connectivity, suggests that initial unaided effort promoted durable memory traces, enabling more effective reactivation even when LLM tools were introduced later.
3. **Metacognitive engagement is higher in the Brain-to-LLM group.**
   Brain-only group might have mentally compared their past unaided efforts with tool-generated suggestions (as supported by their comments during the interviews), engaging in self-reflection and elaborative rehearsal, a process linked to executive control and semantic integration, as seen in their EEG profile.

The significant gap in quoting accuracy between reassigned LLM and Brain-only groups was not merely a behavioral artifact; it is mirrored in the structure and strength of their neural connectivity. The LLM-to-Brain group's early dependence on LLM tools appeared to have impaired long-term semantic retention and contextual memory, limiting their ability to reconstruct content without assistance. In contrast, Brain-to-LLM participants could leverage tools more strategically, resulting in stronger performance and more cohesive neural signatures.

This next finding should be considered preliminary, as a larger participant sample is needed to confirm the claim (see Limitations section below).

Perhaps one of the more concerning findings is that participants in the LLM-to-Brain group repeatedly focused on a narrower set of ideas, as evidenced by n-gram analysis (see topics COURAGE, FORETHOUGHT, and PERFECT in Figures 82, 83, and 85, respectively) and supported by interview responses. This repetition suggests that many participants may not have engaged deeply with the topics or critically examined the material provided by the LLM.

When individuals fail to critically engage with a subject, their writing might become biased and superficial. This pattern reflects the accumulation of *cognitive debt,* a condition in which repeated reliance on external systems like LLMs replaces the effortful cognitive processes required for independent thinking.

Cognitive debt defers mental effort in the short term but results in long-term costs, such as diminished critical inquiry, increased vulnerability to manipulation, decreased creativity. When participants reproduce suggestions without evaluating their accuracy or relevance, they not only forfeit ownership of the ideas but also risk internalizing shallow or biased perspectives.

Taken together, these findings support an educational model that delays AI integration until learners have engaged in sufficient self-driven cognitive effort. Such an approach may promote both immediate tool efficacy and lasting cognitive autonomy.

Limitations and Future Work

In this study we had a limited number of participants recruited from a specific geographical area, several large academic institutions, located very close to each other. For future work it will be important to include a larger number of participants coming with diverse backgrounds like professionals in different areas, age groups, as well as ensuring that the study is more gender balanced.

This study was performed using ChatGPT, and though we do not believe that as of the time of this paper publication in June 2025, there are any significant breakthroughs in any of the commercially available models to grant a significantly different result, we cannot directly generalize the obtained results to other LLM models. Thus, for future work it will be important to include several LLMs and/or offer users a choice to use their preferred one, if any.

Future work may also include the use of LLMs with other modalities beyond the text, like audio modality.

We did not divide our essay writing task into subtasks like idea generation, writing, and so on, which is often done in prior work [76, 115]. This labeling can be useful to understand what happens at each stage of essay writing and have more in-depth analysis.

In our current EEG analysis we focused on reporting connectivity patterns without examining spectral power changes, which could provide additional insights into neural efficiency. EEG's spatial resolution limits precise localization of deep cortical or subcortical contributors (e.g. hippocampus), thus fMRI use is the next step for our future work.

Our findings are context-dependent and are focused on writing an essay in an educational setting and may not generalize across tasks.

Future studies should also consider exploring longitudinal impacts of tool usage on memory retention, creativity, and writing fluency. Several recent reports highlight absence of ownership of generated content at 16%, similar to this study's finding in Session 1 [129], deskilling happening in 3 months [130], AI 'memory gap' phenomena [131] and multiple papers discussing homogenization of language when using LLM chatbots [132, 133].

As datasets become increasingly contaminated with AI-generated content [116], and as the boundary between human thought and generative AI becomes more difficult to discern [117], future research should prioritize collecting writing samples produced without LLM assistance. This would enable the development of a 'fingerprinted' representation of each participant's general and domain-specific writing style [118, 119], which could be used to predict whether a given text was authored by a particular individual rather than generated by an LLM. In this study, conducted across multiple topics in a group setting, the evidence for detecting LLM-generated essays is more than tangential when assessed within-group; however, the precision of this detection remains limited due to the small sample size.

Energy Cost of Interaction

Though the focus of our paper is the cognitive 'cost' of using LLM/Search Engine in a specific task, and more specifically, the cognitive debt one might start to accumulate when using an LLM, we actually argue that the cognitive cost is not the only concern, material and environmental cost is as high. According to a 2023 study [120] LLM query consumes around 10 times more energy than a search query. It is important to note that this energy does not come free, and it is more likely that the average consumer will be indirectly paying for it very soon [121, 122].

Group	Energy per Query	Queries in 20 Hours	Total Energy (Wh)
LLM	0.3 Wh	600	180
Search Engine	0.03 Wh	600	18

Table 4. Approximate breakdown of energy requirement per hour of LLM (ChatGPT) and Search Engine (Google) based on [120], as well as our very approximate estimates on the total energy impact by the LLM group and Search Engine group.

Conclusions

As we stand at this technological crossroads, it becomes crucial to understand the full spectrum of cognitive consequences associated with LLM integration in educational and informational contexts. While these tools offer unprecedented opportunities for enhancing learning and information access, their potential impact on cognitive development, critical thinking, and intellectual independence demands a very careful consideration and continued research.

The LLM undeniably reduced the friction involved in answering participants' questions compared to the Search Engine. However, this convenience came at a cognitive cost, diminishing users' inclination to critically evaluate the LLM's output or 'opinions' (probabilistic answers based on the training datasets). This highlights a concerning evolution of the 'echo chamber' effect: rather than disappearing, it has adapted to shape user exposure through algorithmically curated content. What is ranked as 'top' is ultimately influenced by the priorities of the LLM's shareholders [123, 125].

Only a few participants in the interviews mentioned that they did not follow the 'thinking' [124] aspect of the LLMs and pursued their line of ideation and thinking.

Regarding ethical considerations, participants who were in the Brain-only group reported higher satisfaction and demonstrated higher brain connectivity, compared to other groups. Essays written with the help of LLM carried a lesser significance or value to the participants (impaired ownership, Figure 8), as they spent less time on writing (Figure 33), and mostly failed to provide a quote from their essays (Session 1, Figure 6, Figure 7).

Human teachers 'closed the loop' by detecting the LLM-generated essays, as they recognized the conventional structure and homogeneity of the delivered points for each essay within the topic and group.

We believe that the longitudinal studies are needed in order to understand the long-term impact of the LLMs on the human brain, before LLMs are recognized as something that is net positive for the humans.

Acknowledgments

We would like to thank Janet Baker for her insightful feedback on the first draft of the manuscript. We also would like to thank Lendra Hassman and Luisa Heiss for their thorough grading of the essays. Finally, we would like to thank all our participants.

Author Contributions

The study was proposed, designed, and executed by NK. NK also covered roughly ¼ of all data recording sessions with the participants. NK and EH processed and analyzed both EEG and

NLP data in this study. NK and EH drafted the manuscript. AVB, YTY, XHL were the interns of NK, who helped with the ¾ of data recording sessions with the participants. JS and IB helped with the state of the art drafting section of the paper. IB additionally supported audio-to-text transcriptions of the participants' interviews. PM gave feedback on the study design and the early draft of the manuscript.

Conflict of Interest

At the time of this publication (June 2025), Dr. Kosmyna holds a Visiting Researcher position at Google. All work related to this project was conducted and completed prior to Dr. Kosmyna's affiliation with Google. The remaining authors declare no conflicts of interest.

Funding

This research did not receive any external funding.